NEET-SS / FET / INI-SS HIGH-YIELD NOTES

Hand & Upper Limb

1. flexor and extensor tendons
2. nerve compression and injury
3. brachial plexus
4. carpal instability and scaphoid
5. hand infections
6. Dupuytren and rheumatoid hand
7. congenital hand

Flexor and extensor tendons

1. Flexor tendon anatomy

- **FDS** splits at the proximal phalanx (**Camper's chiasma**) and inserts on the **middle phalanx**; **FDP** passes through the split to the **distal phalanx base**. **FDP of index acts independently** (separate muscle belly); middle/ring/little FDP share a common belly (**quadriga effect**).
- **FDS to the little finger is absent or weak in about 15-20%**.
- **Flexor sheath pulleys: five annular (A1-A5)** and **three cruciform (C1-C3)**; **A1, A3, A5 over joints (MCPJ, PIPJ, DIPJ)**; **A2 (proximal phalanx) and A4 (middle phalanx)** are **the most important biomechanically** (prevent **bowstringing**). **Thumb:** A1, **oblique pulley (most important)**, A2.
- **Nutrition: synovial diffusion** (major within the sheath) and **vincula** (**vinculum longum and breve**, carry blood via transverse digital arteries); **watershed avascular zones** on the volar aspect of the tendons.

2. Flexor tendon zones (Verdan/Kleinert)

Zone	Extent	Notes
I	Distal to FDS insertion (FDP only)	**Jersey finger** (FDP avulsion)
II	**A1 pulley (distal palmar crease) to FDS insertion**	**"No man's land"** (Bunnell): **both FDS and FDP within the tight fibro-osseous sheath**; adhesions; historically poor results
III	Distal edge of carpal tunnel to A1 pulley	**Lumbrical origins** (palm); better prognosis
IV	**Carpal tunnel**	Median nerve injury associated
V	Proximal to carpal tunnel (forearm)	Multiple tendons, nerves and arteries ("**spaghetti wrist**")
Thumb T I-T V	T I distal to IPJ; **T II A1 to IPJ**; T III thenar eminence; T IV carpal tunnel; T V forearm	FPL

3. Flexor tendon repair

- **Timing: primary repair (within 24 hours)** or **delayed primary (within about 10-14 days)**; beyond **3-4 weeks**: tendon retraction and myostatic contracture: consider **staged reconstruction**.
- **Principles:** atraumatic handling, **multi-strand core suture** (**strength is proportional to the number of strands crossing the repair site**: **4-strand or 6-strand core** (e.g. **Adelaide, cruciate, modified Kessler/Strickland**) allows **early active motion**), **core purchase about 7-10 mm from the cut end**, **dorsally placed core suture** is stronger, **epitendinous running suture** (adds about 10-50% strength and reduces gap/gliding resistance), **locking loops** add strength.
- **Gap formation over 3 mm** at the repair: increased **rupture and adhesions**.
- **Repair strength:** decreases between **days 5 and 21** (softening phase) and recovers after that.
- **Pulleys:** preserve **A2 and A4**; **venting (partial release) of A2/A4 (up to about two-thirds of A2 or the whole A4 if needed)** allows passage of bulky repairs without significant bowstringing.
- **FDS in zone II:** repair both slips if possible; **repair only FDP (excise one slip of FDS)** when the canal is too tight.
- **Rehabilitation:**
  - **Early active motion** (**Saint John protocol, Manchester short splint**, true active flexion with 4+ strand repairs): **best range of motion**; slightly higher rupture than passive regimens in some studies.
  - **Kleinert (active extension, passive flexion with rubber bands)**, **Duran (controlled passive motion)**.
  - **Dorsal blocking splint**: **wrist neutral to slight flexion (0-20 degrees), MCPJ flexed about 50-70 degrees, IPJs extended**, for about **4-6 weeks**.
  - Strengthening after about 8-12 weeks.
- **Complications: adhesions** (commonest; tenolysis after **3-6 months** once soft tissues settle and passive ROM is full), **rupture (about 4-10%, usually days 7-14; re-repair if early)**, **joint contracture (PIPJ flexion)**,

bowstringing, **trigger finger**, **quadriga** (over-advanced FDP repair restricts the other fingers' flexion), **lumbrical-plus finger** (paradoxical PIPJ extension on attempted flexion when FDP is cut distal to the lumbrical origin or the graft is too long: lumbrical pulls the extensor apparatus).

4. Jersey finger (FDP avulsion; zone I)

- **Ring finger** commonest (about 75%): forced extension of an actively flexing finger (grabbing a jersey).
- **Leddy and Packer classification:**
  - **Type I:** tendon retracts into the **palm** (both vincula ruptured): **repair within 7-10 days** (poor vascularity, rapid myostatic contracture).
  - **Type II:** retracts to the **PIPJ** (long vinculum intact): repair within about **6 weeks** (commonest type).
  - **Type III: large bony fragment caught at the A4 pulley** (retracts to DIPJ): **ORIF**.
  - **Type IV:** bony fragment + tendon avulsed from the fragment: fix both.
  - **(Type V: with distal phalanx fracture.)**
- **Repair:** pull-out button, **suture anchors**.
- **Late presentation:** DIPJ arthrodesis/tenodesis, accept the deficit (FDS still flexes the PIPJ), **two-stage grafting** (rarely).

5. Flexor tendon reconstruction

- **Single-stage free tendon graft** (palmaris longus: absent in about 15%; plantaris; toe extensors) if the bed is good and pulleys intact.
- **Two-stage (Hunter) reconstruction:** stage 1: **silicone rod** inserted to form a **pseudosheath** (and pulley reconstruction), **stage 2 (about 3 months)**: tendon graft through the new sheath. For **scarred beds, damaged pulleys**.
- **Boyes preoperative classification:** grade 1 good (minimal scar) to grade 5 salvage (multiple digits, poor prognosis).
- **Pulley reconstruction:** tendon graft looped around the phalanx (**triple loop for A2**), **under the extensor mechanism for A2, over it for A4** (around the bone).

6. Extensor tendon anatomy

- **Six dorsal wrist compartments (radial to ulnar):**
  - **APL + EPB** (**De Quervain's tenosynovitis**; **separate EPB subsheath** in many: release it too),
  - **ECRL + ECRB** (**intersection syndrome**: proximal, where compartment 1 muscles cross compartment 2: about **4-8 cm proximal to Lister's tubercle**),
  - **EPL** (ulnar to **Lister's tubercle**: **attrition rupture after undisplaced distal radius fractures**, typically 4-8 weeks later),
  - **EDC + EIP** (**posterior interosseous nerve** on the floor),
  - **EDM** (overlies the DRUJ),
  - **ECU** (subsheath; ECU subluxation).
- **Juncturae tendinum** (between EDC tendons, dorsum of the hand): mask proximal EDC lacerations (finger can still extend).
- **EIP and EDM lie ulnar to the EDC tendons** of their digit (independent extension); **EIP used for EPL reconstruction** (tendon transfer).
- **Extensor mechanism of the finger:**
  - **Central slip** (EDC) inserts on the **base of the middle phalanx** (extends the PIPJ),
  - **Lateral bands** (from **interossei and lumbricals** plus EDC): join to form the **terminal tendon** at the distal phalanx (extends DIPJ),
  - **Sagittal bands** (centralise the EDC over the MCPJ; extend the MCPJ via a sling on the proximal phalanx),
  - **Triangular ligament** (dorsally, between the lateral bands: prevents volar subluxation),
  - **Transverse retinacular ligament** (prevents dorsal subluxation of lateral bands),
  - **Oblique retinacular ligament (Landsmeer):** volar to the PIPJ axis, dorsal to DIPJ: **links PIPJ and DIPJ extension** (tight in boutonniere deformity).
- **Intrinsics: flex the MCPJ and extend the IPJs** (lumbricals originate from FDP: **the only muscles that relax their own antagonist**).

7. Extensor tendon zones (Verdan)

Zone	Location	Classic injury
I	**DIPJ**	**Mallet finger**
II	Middle phalanx	Lateral band / terminal tendon lacerations
III	**PIPJ**	**Boutonniere (central slip)**
IV	Proximal phalanx	Partial lacerations (broad tendon)
V	**MCPJ**	**Fight bite (human bite)**; **sagittal band rupture** ("**boxer's knuckle**")
VI	Metacarpal	Lacerations proximal to the juncturae
VII	**Wrist (extensor retinaculum)**	Retinacular adhesions
VIII	Distal forearm	Musculotendinous junction
IX	Proximal forearm (muscle bellies)	

- **Odd zones over joints** (I, III, V, VII); thumb T I (IPJ) to T V.

8. Mallet finger (zone I)

- **Forced flexion of an extended DIPJ**: terminal tendon rupture or bony avulsion of the dorsal base of the distal phalanx. **Untreated: swan-neck deformity** (extensor imbalance: PIPJ hyperextension).
- **Doyle classification:**
  - **Type I:** closed, ± small avulsion fracture (commonest),
  - **Type II:** laceration at/proximal to the DIPJ,
  - **Type III:** deep abrasion with loss of skin and tendon,
  - **Type IV:** mallet fracture: **A** transepiphyseal (children: **Seymour fracture** if open with nail bed injury: **requires irrigation, debridement, nail-bed repair, antibiotics**: risk of osteomyelitis); **B** hyperflexion injury with fracture **20-50% of the articular surface**; **C** hyperextension with **over 50% articular surface** and **volar subluxation of the distal phalanx**.

- **Treatment:**
  - **Most closed mallet injuries (tendinous or bony): continuous extension splinting of the DIPJ (slight hyperextension) for 6-8 weeks full-time**, then night splint for 2-4 weeks (**PIPJ left free**); **any flexion during this time restarts the clock**. Effective even if presentation is delayed (up to about 3 months).
  - **Surgery (relative): volar subluxation of the distal phalanx** and/or **large fragment (over about 30-50% of the articular surface)**: **extension block pinning (Ishiguro)**, ORIF. Open injuries: repair.
  - **Chronic with swan neck: central slip tenotomy (Fowler)**, **spiral oblique retinacular ligament reconstruction (SORL)**, DIPJ arthrodesis.

9. Boutonniere deformity (zone III)

- **Central slip rupture**: PIPJ flexion; lateral bands migrate **volar to the PIPJ axis** (triangular ligament attenuates): become **PIPJ flexors** and **hyperextend the DIPJ**.
- **Acute injury: Elson test** (**most reliable**: PIPJ flexed 90 degrees over the table edge; attempt to extend the middle phalanx against resistance: **with central slip rupture, the DIPJ goes rigid/extends** (lateral bands act); with an intact slip, the DIPJ remains floppy).
- **Treatment:**
  - **Closed acute: PIPJ extension splinting for 6 weeks, DIPJ free (active DIPJ flexion exercises to pull the lateral bands dorsally)**.
  - **Open laceration/large avulsion fracture:** repair/fix.
  - **Chronic: supple: splinting/serial casting first** (most improve); surgical reconstruction (**Fowler/Dolphin terminal tendon tenotomy** for DIPJ hyperextension; central slip reconstruction) only once **passive PIPJ extension is restored**. **Pseudoboutonniere** = PIPJ hyperextension injury with volar plate scarring (flexion contracture without DIPJ hyperextension).

10. Sagittal band rupture ("boxer's knuckle", zone V)

- Usually **radial sagittal band** (middle finger commonest): **ulnar subluxation of EDC** with MCPJ flexion; extension lag at the MCPJ.
- **Acute: sagittal band (relative motion/yoke) splint** for about 6 weeks; **chronic/failed:** surgical repair or **centralisation (sagittal band reconstruction)**.

11. Fight bite (clenched-fist injury, zone V)

- Small wound over the MCPJ: **assume joint penetration and tendon injury**; skin, tendon and capsule wounds move apart when the fingers are extended.
- **Organisms: mixed: *Streptococcus*, *S. aureus*, *Eikenella corrodens* (classic), anaerobes**: **co-amoxiclav (amoxicillin-clavulanate)**; *Eikenella* is resistant to first-generation cephalosporins/clindamycin.
- **Exploration, irrigation and debridement**, leave wound open, splint; tendon repair delayed if contaminated.

12. Other extensor problems

- **EPL rupture after distal radius fracture** (often undisplaced): **EIP to EPL transfer** (commonest) or free graft.
- **Vaughan-Jackson syndrome** (rheumatoid arthritis): **attrition rupture of the ulnar extensor tendons (EDM, then EDC to little and ring) over a dorsally prominent ulnar head (caput ulnae)**: **tenosynovectomy, Darrach/distal ulna procedure, tendon transfers (EIP or side-to-side)**. **Mannerfelt-Norman lesion**: **FPL attrition rupture over a scaphoid osteophyte** (rheumatoid).
- **Extensor repair: zones IV-VIII**: core sutures (thinner tendons in zones I-IV need running/horizontal mattress or figure-of-eight sutures); **relative motion extension splinting (yoke)** for zones IV-VII allows early active motion; **static or dynamic splinting** otherwise.

Exam pearl: **Zone II (A1 pulley to FDS insertion) = no man's land. A2 and A4 pulleys prevent bowstringing; thumb oblique pulley most important. Repair strength depends on the number of core strands; 4-6 strand + epitendinous allows early active motion. Weakest days 5-21; ruptures usually days 7-14. Adhesions = commonest complication.**

Exam pearl: **Jersey finger: ring finger; Leddy-Packer I (palm, both vincula) repair within 7-10 days; III bony fragment caught at A4. Lumbrical-plus finger: paradoxical PIPJ extension. Quadriga: over-advanced FDP.**

Exam pearl: **Mallet: 6-8 weeks continuous DIPJ extension splint (PIPJ free); fix if volar subluxation. Boutonniere: Elson test; PIPJ splinted in extension, DIPJ free to flex. Fight bite: Eikenella, co-amoxiclav, explore. EPL rupture after undisplaced distal radius fracture: EIP transfer. Vaughan-Jackson = ulnar extensors over the ulnar head (RA); Mannerfelt = FPL over the scaphoid.**

Nerve compression and injury

1. Nerve injury basics

Structure

- **Endoneurium** (around each axon), **perineurium** (around a fascicle: the **blood-nerve barrier** and main tensile strength), **epineurium** (around the whole nerve, plus between fascicles).
- **Myelinated fibres:** saltatory conduction at the nodes of Ranvier; **Schwann cells** (PNS).

Seddon and Sunderland classifications

Seddon	Sunderland	Damage	Tinel sign	Recovery
Neurapraxia	**I**	**Local conduction block (demyelination)**, axon intact	**Absent (no advancing Tinel)**	**Complete, within days to about 12 weeks**
Axonotmesis	**II**	**Axon disrupted, endoneurium intact**	**Advancing**	**Complete** (Wallerian degeneration, regrowth along intact tubes)
	III	Axon + **endoneurium** disrupted, perineurium intact	Advancing	**Incomplete/variable** (mismatching, scarring)
	IV	Axon, endoneurium + **perineurium** disrupted; **epineurium intact** (neuroma-in-continuity)	Static (does not advance)	**None without surgery** (excise and repair/graft)

Seddon	Sunderland	Damage	Tinel sign	Recovery
Neurotmesis	**V**	**Complete transection**	Static	**None without surgery**
	VI (Mackinnon)	**Mixed** pattern		Variable

- **Wallerian degeneration:** the distal segment degenerates (**begins within 24-48 hours**); macrophages and Schwann cells clear debris; Schwann cells form **bands of Büngner** to guide regeneration.
- **Regeneration rate:** about **1 mm/day (about 1 inch/month)** after a latent period; **advancing Tinel sign** tracks it.
- **Motor end plates degenerate irreversibly after about 12-18 months** of denervation: **motor recovery requires reinnervation before this**: hence **nerve transfers** close to the target for proximal injuries. Sensory receptors survive longer.
- **Prognosis is better with: young age**, **distal (closer to the target) injuries**, **pure motor or pure sensory nerves**, **sharp clean injuries**, **early repair**, **short gaps**; **worse with:** old age, proximal injuries, mixed nerves, crush/avulsion, delay, smoking/diabetes.

Investigations

- **Nerve conduction studies (NCS) and EMG:** wait **about 3 weeks** after injury (Wallerian degeneration must occur before denervation changes appear). **Fibrillation potentials and positive sharp waves** = denervation (axonal injury); **motor unit potentials (nascent, polyphasic)** = reinnervation (appear before clinical recovery).
- **Sensory nerve action potentials (SNAPs) preserved with numb skin = preganglionic (root avulsion) lesion** (the dorsal root ganglion cell body is intact) (brachial plexus).
- **Ultrasound/MRI neurography:** continuity, neuroma.

Repair principles

- **Primary repair (sharp injuries)** within days; **tension-free**; **epineurial repair** (most common; align with surface vessels, fascicular pattern),

group fascicular repair where fascicles are topographically defined (e.g. motor branch of the ulnar nerve at the wrist).

- **Gaps: nerve conduit** (hollow tube; **gaps under about 3 cm in small-calibre sensory nerves**), **processed nerve allograft** (gaps up to about 5-7 cm in sensory nerves), **autograft (sural nerve: gold standard for larger gaps)**. Avoid tension (**over about 10% elongation impairs blood flow**).
- **Nerve transfers** (e.g. **AIN to deep motor branch of ulnar nerve** (supercharge end-to-side), **Oberlin transfer** (ulnar fascicle to biceps branch), **spinal accessory to suprascapular**, **radial nerve triceps branch to axillary nerve**): convert a proximal injury into a distal one.
- **Tendon transfers** for late or irrecoverable motor loss.
- **Closed injuries with fractures (e.g. radial nerve with humeral shaft fracture):** most are neurapraxia/axonotmesis: **observe** with serial examination and EMG at about 3 months; **explore** if **no recovery by about 3-6 months**, open fractures, **nerve palsy after manipulation (relative)**, penetrating trauma.

2. Carpal tunnel syndrome (CTS)

- **Commonest compressive neuropathy.** Carpal tunnel boundaries: **floor**: carpal bones; **roof**: **transverse carpal ligament (flexor retinaculum)** from **scaphoid tubercle and trapezium** (radial) to **pisiform and hook of hamate** (ulnar). Contents: **median nerve + 9 flexor tendons (4 FDS, 4 FDP, FPL)**.
- **Associations: female**, **pregnancy**, **diabetes**, **hypothyroidism**, **rheumatoid arthritis**, **obesity**, **acromegaly**, **amyloidosis** (dialysis, **transthyretin amyloid: carpal tunnel often precedes cardiac amyloidosis**), **gout**, mass lesions, **distal radius fracture (acute CTS: urgent release if progressive)**.
- **Symptoms: nocturnal paraesthesia** in the radial three and a half digits (**palm spared**: **palmar cutaneous branch** arises about **5-6 cm proximal to the wrist crease and passes superficial to the retinaculum**), clumsiness, **thenar wasting (APB)** in severe cases.
- **Tests:**
  - **Durkan (carpal compression) test** (**most sensitive**),

  - **Phalen test** (wrist flexion 60 s),
  - Tinel sign,
  - **Semmes-Weinstein monofilament** (**most sensitive threshold test for early compression**), **two-point discrimination** (innervation density: **late** change).
  - **CTS-6** clinical diagnostic tool.
- **NCS:** prolonged **distal sensory latency** (earliest), motor latency (**over 4.5 ms**), reduced amplitudes (axonal loss); **EMG** fibrillations in APB (severe).
- **Treatment:**
  - **Night splinting in neutral**, **corticosteroid injection** (also diagnostic: **good response predicts a good surgical outcome**), treat cause.
  - **Carpal tunnel release (open or endoscopic):** similar long-term results; endoscopic: faster return to work, **higher risk of transient nerve injury**. Release **in line with the radial border of the ring finger** (avoids the **recurrent motor branch** and the **palmar cutaneous branch**; **Kaplan cardinal line**).
  - **Recurrent motor branch variants: extraligamentous recurrent (commonest, about 50%)**, **subligamentous**, **transligamentous (about 25%; at risk)**.
  - **Pillar pain** (commonest complaint after release), scar tenderness, **incomplete release** (commonest cause of persistent symptoms), recurrence, **CRPS**, nerve injury.

3. Pronator syndrome and AIN syndrome

- **Pronator syndrome:** median nerve compression in the **proximal forearm**: **sensory symptoms including the palm (palmar cutaneous branch territory)**, **no nocturnal symptoms**, **NCS usually normal**. **Compression sites (proximal to distal):**
  - **Ligament of Struthers** (from a **supracondylar process** of the humerus, about 5 cm above the medial epicondyle),
  - **Lacertus fibrosus (bicipital aponeurosis)**,
  - **Between the two heads of pronator teres** (**commonest**),
  - **FDS arch (sublimis bridge)**.

  - **Provocative tests:** resisted pronation with elbow extension (pronator teres), resisted elbow flexion with supination (lacertus), resisted middle finger PIPJ flexion (FDS arch).
- **Anterior interosseous nerve (AIN) syndrome: pure motor**: **FPL, FDP to index (and middle), pronator quadratus**; **cannot make an "OK" sign (pinch sign: flat pulp-to-pulp pinch)**; **no sensory loss**. Causes: compression (same sites), **Parsonage-Turner syndrome (brachial neuritis: often the true cause: severe shoulder pain preceding weakness)**, **supracondylar fracture in children (AIN is the commonest nerve injured in extension-type fractures)**. Observe for 3-6 months (most recover); explore if no recovery.

4. Cubital tunnel syndrome (ulnar nerve at the elbow)

- **Second commonest compressive neuropathy. Sites of compression (proximal to distal):**
  - **Arcade of Struthers** (about 8 cm proximal to the medial epicondyle; medial intermuscular septum),
  - **Medial intermuscular septum**,
  - **Medial epicondyle** (subluxation, valgus deformity: **tardy ulnar nerve palsy after lateral condyle non-union with cubitus valgus**),
  - **Osborne's ligament (cubital tunnel retinaculum: roof of the cubital tunnel)** (**commonest site**),
  - **Between the two heads of FCU (Osborne's fascia)**,
  - **Deep flexor-pronator aponeurosis**.
- **Elbow flexion increases pressure and stretches the nerve** (nerve elongates about 4.5-8 mm).
- **Symptoms/signs:** paraesthesia of the **little finger and ulnar half of the ring finger** + **dorsoulnar hand (dorsal cutaneous branch: distinguishes from Guyon canal compression)**, **intrinsic weakness**, **Froment sign** (FPL substitutes for **adductor pollicis**: IPJ flexion during key pinch), **Jeanne sign** (MCPJ hyperextension of the thumb), **Wartenberg sign** (**abducted little finger: unopposed EDM (radial nerve), weak third palmar interosseous**), **claw hand (ulnar claw: ring and little fingers)**, **elbow flexion test**, Tinel.

- **Ulnar paradox: proximal (elbow) lesions cause less clawing than distal (wrist) lesions** because **FDP to ring and little fingers is also paralysed** in high lesions.
- **Treatment:** activity modification, **night splint in extension (about 45 degrees)**; surgery: **in situ decompression (open or endoscopic: first-line; equivalent outcomes to transposition, fewer complications)**, **anterior transposition (subcutaneous or submuscular)** for **unstable/subluxing nerves** or after failed decompression, **medial epicondylectomy**. **Medial antebrachial cutaneous nerve** branches at risk (painful neuroma).
- **McGowan classification** (grade I sensory, II weakness, III atrophy/paralysis): severe/long-standing disease has poorer recovery.

5. Guyon canal (ulnar tunnel) syndrome

- **Boundaries: roof: volar carpal ligament (and palmaris brevis)**, **floor: transverse carpal ligament (and pisohamate/pisometacarpal ligaments)**, **radial wall: hook of hamate**, **ulnar wall: pisiform**.
- **Zones (Gross and Gelberman):**
  - **Zone 1** (proximal to the bifurcation): **motor and sensory deficit**,
  - **Zone 2** (deep motor branch, around the hook of hamate): **pure motor**,
  - **Zone 3** (superficial sensory branch): **pure sensory**.
- **Causes: ganglion (commonest)**, **hook of hamate fracture/non-union** (golfers, baseball, cyclists' palsy), **ulnar artery thrombosis/aneurysm (hypothenar hammer syndrome)**, anomalous muscles, lipoma.
- **Dorsal cutaneous branch spared** (arises **about 5-8 cm proximal to the wrist**): **dorsal hand sensation normal** (vs cubital tunnel). **FDP to ring/little and FCU normal**.
- **Treatment:** CT/MRI to identify the cause; decompression, excision of ganglion/hamate hook.

6. Radial nerve compression

- **Radial tunnel syndrome: pain** (lateral forearm, **about 4-5 cm distal to the lateral epicondyle, over the radial tunnel**), **no motor weakness**; mimics **tennis elbow** (coexist in about 5%); **pain on resisted middle**

finger extension and resisted supination; NCS normal. **Treatment:** non-operative first; decompression.

- **Posterior interosseous nerve (PIN) syndrome: motor weakness without sensory loss**: **finger and thumb extension drop**, **wrist extends in radial deviation** (ECRL (radial nerve proper) preserved; **ECU paralysed**).
- **Compression sites in the radial tunnel (FREAS):**
  - **Fibrous bands** anterior to the radiocapitellar joint,
  - **Recurrent radial vessels ("leash of Henry")**,
  - **ECRB tendinous margin**,
  - **Arcade of Frohse (proximal edge of the supinator: commonest site)**,
  - **Supinator distal edge**.
- **Other causes: Monteggia fracture-dislocation (PIN palsy: commonest nerve injury, usually neurapraxia)**, **lipoma**, **rheumatoid synovitis of the elbow** (radiocapitellar), ganglion. **Differentiate from extensor tendon rupture** (RA): **tenodesis test** (wrist flexion extends the MCPJs if tendons are intact).
- **Wartenberg syndrome (cheiralgia paraesthetica): superficial radial nerve** compression **between brachioradialis and ECRL** (tight watch, handcuffs): dorsoradial hand paraesthesia; **Tinel**, pronation with wrist ulnar deviation provokes it.
- **Saturday night palsy** (radial nerve in the **spiral groove**): **wrist drop**, **triceps spared** (branches arise proximal).

7. Other compressive and nerve lesions

- **Thoracic outlet syndrome: neurogenic (commonest, about 95%; lower trunk C8-T1: ulnar-sided symptoms, Gilliatt-Sumner hand: wasting of the thenar eminence (APB))**, **venous (Paget-Schroetter: effort thrombosis of the subclavian vein)**, **arterial (cervical rib: subclavian artery aneurysm, emboli)**. **Causes:** cervical rib, anomalous bands, scalene hypertrophy. **Tests: Roos (elevated arm stress) test**, **Adson test**, Wright. Physiotherapy first; first rib resection/scalenectomy.
- **Suprascapular nerve:**

    - **Suprascapular notch** (under the **transverse scapular ligament**): **supraspinatus + infraspinatus** weakness,
    - **Spinoglenoid notch** (**paralabral cyst** from a **SLAP tear**): **isolated infraspinatus** weakness (painless wasting: volleyball players).
- **Quadrilateral space syndrome: axillary nerve + posterior circumflex humeral artery** (fibrous bands; throwers): posterior shoulder pain, **teres minor** atrophy on MRI.
- **Long thoracic nerve palsy: medial scapular winging** (serratus anterior). **Spinal accessory nerve palsy** (posterior triangle lymph node biopsy): **lateral winging** (trapezius) with shoulder drooping.
- **Double-crush syndrome:** proximal compression (cervical radiculopathy) makes the distal nerve more susceptible to compression.

8. Tendon transfers (for nerve palsies)

- **Principles: adequate strength (lose one MRC grade after transfer: muscle of grade 5 needed)**, **expendable donor**, **similar excursion** (**wrist flexors about 33 mm, finger extensors about 50 mm, finger flexors about 70 mm**: **"3-5-7" rule**), **straight line of pull**, **one tendon one function**, **synergistic transfers easier to retrain** (wrist flexion with finger extension), **supple joints, stable soft-tissue bed**, **tissue equilibrium** (wait for swelling/scar to settle).
- **Radial nerve palsy (high): pronator teres to ECRB** (wrist extension), **FCR or FCU to EDC** (finger extension), **palmaris longus to EPL (rerouted)** (thumb extension). **Early "internal splint" (PT to ECRB)** while awaiting nerve recovery.
- **Low median nerve palsy (opposition): opponensplasty**: **EIP (Burkhalter)**, **FDS ring (Bunnell)**, **palmaris longus (Camitz: for severe CTS; mainly abduction)**, **abductor digiti minimi (Huber: congenital thumb hypoplasia)**.
- **High median:** + **ECRL to FDP index/middle**, **brachioradialis to FPL**.
- **Ulnar nerve palsy: clawing correction**: **Zancolli capsulodesis** (MCPJ volar plate), **Stiles-Bunnell (FDS lasso)**, **ECRL/ECRB transfer (Brand) to the lateral bands**; **key pinch**: **ECRB to adductor pollicis (Smith)**, **FDS ring to adductor**; **index abduction**: **EIP or APL slip to first dorsal**

interosseous; **Bouvier manoeuvre** (blocking MCPJ hyperextension corrects the claw: **intact extensor mechanism**: **simple (static) procedure suffices** if positive).

Exam pearl: **Seddon/Sunderland: neurapraxia (I) no Tinel, full recovery; axonotmesis II full, III/IV incomplete/none; neurotmesis V needs surgery. Regeneration 1 mm/day. Motor end plates lost after 12-18 months: nerve transfers. EMG at 3 weeks. Preserved SNAP with anaesthetic skin = preganglionic.**

Exam pearl: **CTS: palm spared (palmar cutaneous branch); Durkan most sensitive; Semmes-Weinstein for early compression; incomplete release = commonest cause of persistent symptoms; transligamentous motor branch at risk. Pronator syndrome: palm involved, commonest between pronator teres heads. AIN: no OK sign, no sensory loss; consider Parsonage-Turner.**

Exam pearl: **Cubital tunnel: dorsal ulnar sensation lost (vs Guyon: spared); Osborne's ligament commonest; in situ decompression first-line. Ulnar paradox: distal lesions claw more. Guyon zone 2 = pure motor (hook of hamate). PIN palsy: radial deviation of the wrist on extension; arcade of Frohse; RA: tenodesis test excludes tendon rupture.**

Exam pearl: **Tendon transfers: lose one grade; 3-5-7 cm excursion; radial palsy: PT to ECRB, FCR/FCU to EDC, PL to EPL. Opponensplasty: EIP/FDS ring/Camitz (PL)/Huber (ADM).**

Brachial plexus

1. Anatomy

- **Roots C5-T1** (variations: **prefixed C4-C8**, **postfixed C6-T2**). **Mnemonic: Roots, Trunks, Divisions, Cords, Branches** ("Randy Travis Drinks Cold Beer").
- **Roots** (between scalenus anterior and medius): branches **dorsal scapular nerve (C5: rhomboids, levator scapulae)**, **long thoracic nerve (C5-7: serratus anterior)**, contribution to the phrenic nerve (C5).
- **Trunks** (posterior triangle of the neck): **upper (C5-6)**, **middle (C7)**, **lower (C8-T1)**. Upper trunk branches: **suprascapular nerve (supraspinatus, infraspinatus)**, **nerve to subclavius**.
- **Divisions** (behind the clavicle): anterior and posterior; **no branches**.
- **Cords** (named by relation to the **axillary artery**, second part, deep to pectoralis minor):
  - **Lateral (C5-7): lateral pectoral nerve**, **musculocutaneous nerve**, lateral root of the median nerve.
  - **Medial (C8-T1): medial pectoral nerve**, **medial cutaneous nerves of arm and forearm**, **ulnar nerve**, medial root of the median nerve.
  - **Posterior (C5-T1): upper and lower subscapular nerves** (subscapularis, teres major), **thoracodorsal nerve (C6-8: latissimus dorsi)**, **axillary nerve (C5-6)**, **radial nerve (C5-T1)**.
- **Erb's point:** junction of C5 and C6 roots (upper trunk), **about 2-3 cm above the clavicle** behind the sternocleidomastoid.
- **Sympathetic chain: T1 contributes to the stellate ganglion: Horner syndrome (ptosis, miosis, anhidrosis, enophthalmos)** = **T1 preganglionic avulsion**.

2. Levels of injury: preganglionic vs postganglionic

Feature	Preganglionic (root avulsion; proximal to the dorsal root ganglion)	Postganglionic (distal to the DRG)
Recovery potential	**None (no spontaneous recovery, cannot be repaired directly)**	**Possible** (repair/graft, spontaneous)
Horner syndrome	**Present (T1)**	Absent
Proximal muscle weakness: **serratus anterior (long thoracic), rhomboids (dorsal scapular)**, **paraspinals**	**Paralysed** (winging)	**Spared**
Hemidiaphragm paralysis (phrenic, C5)	Possible	Absent
Tinel sign in the neck (supraclavicular)	**Absent**	**Present**
Severe neuropathic pain (burning, deafferentation)	**Common**	Less
SNAP (sensory nerve action potential)	**Preserved (despite anaesthetic skin)** (DRG cell bodies intact, peripheral axon intact)	**Absent**
Imaging (CT myelography/MRI)	**Pseudomeningocele**, absent rootlets	Normal roots
Histamine test (triple response)	Preserved	Absent
Fractures	Transverse process fractures of the cervical vertebrae, first rib fracture	Clavicle fracture

- **C5-C6 are most often postganglionic ruptures** (strong ligamentous attachment (transverse radicular ligaments) in the gutter of the transverse

processes); **C8-T1 are most often avulsed (preganglionic)** (less protected).

- **Avulsion: central (intradural) avulsion** of rootlets; **peripheral avulsion** at the foramen.

3. Adult traumatic brachial plexus injury

- **Mechanism: high-energy traction**: **motorcycle accidents (young men)**: **head/neck forced away from a depressed shoulder** (upper plexus: C5-6) or **arm forced into abduction** (lower plexus: C8-T1).
- **Patterns:**
  - **Upper (C5-6) (Erb):** shoulder abduction/external rotation and elbow flexion lost; **"waiter's tip" posture**.
  - **Extended upper (C5-7):** + wrist and finger extension weakness.
  - **Lower (C8-T1) (Klumpke):** intrinsic hand paralysis, claw hand, ± **Horner** (rare in isolation in adults).
  - **Pan-plexus (C5-T1)**: commonest pattern in adults (**about 50-75%**): flail anaesthetic arm.
- **Associated injuries: subclavian/axillary artery injury**, **scapulothoracic dissociation** (**lateral displacement of the scapula on chest radiograph**, plexus avulsion + vascular injury: **poor prognosis**), **clavicle fracture**, **first rib fracture**, **head and cervical spine injuries**, **shoulder dislocation (infraclavicular plexus and axillary nerve)**.
- **Infraclavicular injuries** (cords/branches: shoulder dislocation, proximal humerus fractures) have **better prognosis** (postganglionic).
- **Investigations:**
  - **Chest radiograph** (hemidiaphragm, rib fractures), **cervical spine**.
  - **MRI** or **CT myelography** (**pseudomeningoceles**: avulsion; CT myelography more sensitive for rootlet avulsion).
  - **NCS/EMG at about 3-4 weeks** (SNAPs, paraspinal denervation: preganglionic).
  - **Intraoperative** nerve stimulation, **somatosensory evoked potentials (SSEP)**, frozen section (ganglion cells: avulsion).

Management

- **Timing:**
  - **Open/sharp injuries: immediate exploration and repair**.
  - **Closed traction injuries with suspected avulsion (Horner sign, pseudomeningoceles): early exploration (within about 3 months)** for nerve transfers/grafts.
  - **Closed injuries with likely postganglionic lesions: observe 3 months**; explore if no clinical/EMG recovery (**ideally within 3-6 months**; **results deteriorate after about 6 months**; **motor end-plate degeneration by 12-18 months**).
  - **Vascular injury:** emergency repair (± plexus exploration).
- **Surgical options:**
  - **Neurolysis** (neuroma-in-continuity with conduction across it).
  - **Nerve grafting (sural nerve)** from viable roots (postganglionic).
  - **Nerve transfers (neurotisation):**
  - **Elbow flexion: Oberlin transfer** (**fascicle of the ulnar nerve to the biceps motor branch**), **double fascicular transfer** (ulnar fascicle to biceps + **median fascicle to brachialis**) (Mackinnon), **intercostal nerves to musculocutaneous nerve**, **spinal accessory to musculocutaneous**, **phrenic nerve**, **contralateral C7**.
  - **Shoulder: spinal accessory (distal) to suprascapular nerve** (via posterior approach preferred), **radial nerve (triceps long-head branch) to axillary nerve (Leechavengvongs/Somsak transfer)**.
  - **Priority in pan-plexus injury: elbow flexion first**, then **shoulder stability/abduction**, then hand sensation, then wrist/finger function.
  - **Free functioning muscle transfer (gracilis)** reinnervated by spinal accessory or intercostal nerves for late presentation or pan-plexus (elbow flexion and/or finger flexion: **Doi double free muscle**).
  - **Tendon transfers** (incomplete injuries, late), **Steindler flexorplasty** (proximal advancement of the flexor-pronator origin for elbow flexion), **latissimus/pectoralis transfer for elbow flexion**, **trapezius transfer for shoulder abduction**.
  - **Shoulder arthrodesis** (flail shoulder with **intact scapulothoracic muscles: trapezius, serratus anterior**): **about 20-30 degrees**

abduction, 20-30 degrees flexion, 40 degrees internal rotation ("salute position"/hand-to-mouth).
    - **Wrist arthrodesis**, **amputation** (above elbow with shoulder fusion: seldom improves function; **does not relieve pain**).
- **Neuropathic (deafferentation) pain:** pain medication (gabapentinoids), **dorsal root entry zone (DREZ) lesioning** for intractable pain in avulsions.

4. Obstetric (birth) brachial plexus palsy (BBPP/NBPP)

- **Incidence about 1-2 per 1,000 live births. Risk factors: shoulder dystocia (strongest)**, **macrosomia (large for gestational age, maternal diabetes)**, **breech delivery** (**bilateral, upper-root avulsions**), instrumental delivery, prolonged labour, previous affected child.
- **Narakas classification:**
    - **Group I: C5-6** (**Erb's palsy**; commonest; **best prognosis: about 90% spontaneous recovery**): **waiter's tip** (adducted, internally rotated shoulder, extended elbow, pronated forearm, flexed wrist).
    - **Group II: C5-7** (extended Erb): + wrist/finger extension loss (wrist drop).
    - **Group III: C5-T1 without Horner** (total palsy, flail arm).
    - **Group IV: C5-T1 with Horner syndrome** (**worst prognosis**).
    - (**Klumpke palsy, isolated C8-T1, is very rare.**)
- **Differential diagnosis (pseudoparalysis): clavicle fracture**, proximal humerus physeal fracture, **septic arthritis/osteomyelitis of the shoulder**, cerebral palsy.
- **Prognosis: recovery of biceps (elbow flexion) by 3 months** is the classic prognostic indicator (Gilbert and Tassin): **absence of biceps function at 3 months suggests poor spontaneous recovery: consider microsurgery (3-9 months)**. **Toronto Test Score** (elbow flexion, elbow extension, wrist, finger and thumb extension at 3 months; **score under 3.5 = surgery**), **cookie test at 9 months** (hand-to-mouth with the arm adducted). **Horner syndrome and total palsy: earlier surgery (about 3 months)**.
- **Management:**
    - **Physiotherapy (passive ROM: shoulder external rotation, abduction; prevent contracture)** from about 2-3 weeks.

  - **Primary microsurgical reconstruction** (neuroma excision + sural grafts, nerve transfers) at **3-9 months** if no biceps recovery.
  - **Assessment scales: Active Movement Scale (AMS)**, **Mallet classification** (shoulder function: abduction, external rotation, hand to neck, hand to back, hand to mouth; grades I-V).
- **Secondary deformities:**
  - **Internal rotation contracture of the shoulder** (**subscapularis/pectoralis major contracture, weak external rotators**): **glenohumeral dysplasia (posterior subluxation, glenoid retroversion, pseudoglenoid formation)**: **Waters classification (types I-VII)** on MRI/ultrasound. Monitor **passive external rotation** (in adduction); **ultrasound screening** in infancy.
  - **Treatment: botulinum toxin to subscapularis/pectoralis**, **open/arthroscopic release + reduction (subscapularis lengthening/release)**, **tendon transfer of latissimus dorsi and teres major to the rotator cuff (L'Episcopo/Hoffer/Sever)** for external rotation and abduction, **humeral external rotation (derotation) osteotomy** (late, with severe glenoid deformity).
  - **Elbow flexion contracture**, **radial head dislocation (posterior)**, **forearm supination deformity** (biceps rerouting, forearm osteotomy), **"trumpet sign"** (shoulder abducts on attempted hand-to-mouth: weak external rotation).

5. Other brachial plexus conditions

- **Parsonage-Turner syndrome (neuralgic amyotrophy, acute brachial neuritis): sudden severe shoulder pain** (often after infection, vaccination, surgery) followed days later by **patchy weakness** (**long thoracic, suprascapular, AIN, axillary nerves** commonly); **pain resolves before weakness**; **hourglass constrictions** of nerve fascicles on high-resolution ultrasound/MRI. **Self-limiting** (recovery over months to years); steroids early; surgery (interfascicular neurolysis) for hourglass constrictions without recovery.
- **Radiation-induced plexopathy** (breast cancer; **myokymia** on EMG; usually **upper plexus**, painless) vs **tumour infiltration (Pancoast: lower**

plexus, painful, Horner).

- **Thoracic outlet syndrome** (see nerve compression notes).
- **Stingers/burners** (transient upper trunk traction or root compression in contact sport): unilateral burning dysaesthesia, transient weakness; return to play once symptoms resolve fully; recurrent: investigate (stenosis).
- **Brachial plexus tumours: schwannoma (commonest)**, **neurofibroma** (NF1, plexiform), **malignant peripheral nerve sheath tumour** (NF1; rapid growth, pain).

6. Examination checklist (rapid)

- **Proximal (preganglionic) clues: rhomboids** (dorsal scapular, C5), **serratus anterior** (long thoracic, C5-7: winging), **paraspinals** (posterior primary rami), **diaphragm** (phrenic), **Horner** (T1).
- **Trunk level: supraspinatus/infraspinatus** (suprascapular nerve: from the upper trunk).
- **Cord level: pectoralis major** (clavicular head: lateral cord (lateral pectoral); sternal head: medial cord (medial pectoral)), **latissimus dorsi** (thoracodorsal: posterior cord).
- **Terminal branches:** axillary (deltoid, regimental badge sensation), musculocutaneous (biceps, lateral forearm sensation), radial, median, ulnar.

Exam pearl: **Preganglionic avulsion signs: Horner (T1), winging (long thoracic), rhomboid paralysis (dorsal scapular), paraspinal denervation, hemidiaphragm, absent Tinel, preserved SNAP with anaesthetic skin, pseudomeningocele. C5-6 rupture (postganglionic); C8-T1 avulse.**

Exam pearl: **Adult closed plexus injury: explore by 3-6 months; motor end plates lost by 12-18 months. Elbow flexion priority: Oberlin (ulnar fascicle to biceps branch); shoulder: spinal accessory to suprascapular + triceps branch to axillary. Late: free functioning gracilis; flail shoulder: arthrodesis.**

Exam pearl: **Obstetric palsy: shoulder dystocia; Narakas I (C5-6) best, IV (with Horner) worst; no biceps recovery at 3 months (Gilbert) or Toronto score under 3.5 = microsurgery at 3-9 months. Secondary internal rotation contracture: glenoid dysplasia (Waters); latissimus/teres major transfer (L'Episcopo/Hoffer).**

Carpal instability and scaphoid

Part A: Scaphoid fractures

1. Anatomy

- **Scaphoid:** the **link between the proximal and distal carpal rows**; **about 80% covered by articular cartilage**; regions: **proximal pole, waist, distal pole, tubercle**.
- **Blood supply (Gelberman): dorsal carpal branch of the radial artery** enters the **dorsal ridge (non-articular) at the waist**, supplying **about 70-80% (including the proximal pole) by retrograde flow**; **superficial palmar branch** supplies the **distal 20-30% (tubercle)**. **Retrograde supply: proximal pole fractures have the highest AVN/non-union risk.**
- **Commonest carpal bone fractured (about 60-70% of carpal fractures)**; **young men**; **FOOSH (wrist dorsiflexed over about 95 degrees and radially deviated)**.

2. Diagnosis

- **Anatomical snuffbox tenderness** (sensitive, not specific), **scaphoid tubercle tenderness**, **pain on axial compression of the thumb**; **combination of signs** improves specificity.
- **Radiographs: scaphoid series (PA, lateral, PA with ulnar deviation (scaphoid view), semi-pronated oblique)**: **up to about 20% of fractures are occult initially**.
- **Clinical suspicion with normal radiographs: immobilise and image**: **MRI (gold standard for occult fractures: sensitivity and specificity near 100%; best within the first days)**, **CT** (displacement, union, humpback), **bone scan** (72 hours; sensitive, not specific). **Early MRI is cost-effective** (avoids unnecessary immobilisation).
- **CT along the long axis of the scaphoid**: assess **displacement** (**over 1 mm**), **humpback deformity** (**intrascaphoid angle over 35 degrees on sagittal CT**; **height-to-length ratio over 0.65**), comminution.

- **Displacement criteria (unstable): over 1 mm step or gap**, **intrascaphoid angle over 35 degrees**, **radiolunate angle over 15 degrees (DISI)**, **comminution**, **associated perilunate injury**, **proximal pole fracture**.

3. Classification

- **Herbert classification:**
  - **A (stable acute): A1 tubercle**, **A2 incomplete waist**,
  - **B (unstable acute): B1 distal oblique**, **B2 complete waist**, **B3 proximal pole**, **B4 trans-scaphoid perilunate fracture-dislocation**, (B5 comminuted),
  - **C: delayed union**, **D: non-union** (D1 fibrous, D2 sclerotic/pseudarthrosis).
- **Russe classification (orientation): horizontal oblique, transverse, vertical oblique (least stable: shear)**.
- **Location: waist (about 65-70%)**, **proximal pole (about 10-20%)**, **distal pole/tubercle (about 10-15%: heal well)**.

4. Management

- **Tubercle and incomplete fractures: short below-elbow cast or splint 4-6 weeks**.
- **Undisplaced waist fractures: cast immobilisation (below-elbow; thumb inclusion not necessary)** for **6-8 weeks** (**union about 90%+**) vs **percutaneous screw fixation** (**faster return to work/sport**, **SWIFFT trial (UK, 2020): no difference in function at 1 year between surgery and cast with early fixation of non-unions**; **cast with CT at 6 weeks and fixation of non-unions is the standard**).
- **Displaced waist fractures (over 1-2 mm): ORIF/percutaneous fixation** (**non-union about 50% with cast**).
- **Proximal pole fractures: fixation (dorsal approach)** (**high non-union risk ~30% with casting**).
- **Fixation: headless compression screw (Herbert/Acutrak)** along the **central axis of the scaphoid** (**central placement = strongest, fastest union**); **volar approach** for waist/distal fractures (**preserves the dorsal**

blood supply), **dorsal approach** for proximal pole fractures (**risk to the dorsal ridge vessels; wrist flexion to access the entry point**).

- **Paediatric scaphoid fractures:** usually **distal third**, heal well with casting.

5. Scaphoid non-union

- **Non-union about 5-10%** overall; **risk factors: displacement**, **proximal pole**, **delay in diagnosis/treatment (over 4 weeks)**, **smoking**, **inadequate immobilisation**, **vertical oblique pattern**, **associated carpal instability**.
- **Natural history: SNAC wrist (scaphoid non-union advanced collapse)**: predictable progression:
  - **Stage I: radial styloid-scaphoid arthritis** (distal scaphoid fragment and radial styloid),
  - **Stage II: + scaphocapitate arthritis**,
  - **Stage III: + capitolunate (midcarpal) arthritis** (**radiolunate joint spared**: **proximal pole moves with the lunate**),
  - (Stage IV: pancarpal/radiolunate).
- **Humpback deformity**: **palmar flexion of the distal fragment + dorsal extension of the proximal fragment (with the lunate): DISI**.
- **Treatment of non-union (no arthritis):**
  - **Stable, undisplaced, fibrous non-union without humpback**: **percutaneous screw ± bone graft**.
  - **Humpback/displaced: open reduction, volar wedge (corticocancellous) graft from the iliac crest (Fisk-Fernandez)**, **screw fixation**.
  - **Proximal pole AVN (no punctate bleeding at surgery: the best indicator of viability; MRI with gadolinium)**: **vascularised bone graft**: **1,2-intercompartmental supraretinacular artery (1,2-ICSRA) pedicled graft (Zaidemberg, dorsal distal radius)**, **medial femoral condyle free vascularised graft (MFC; for humpback with AVN)**, **volar pronator quadratus pedicle graft (Kuhlmann)**.
  - **Matti-Russe** (volar inlay cancellous graft: historical).

- **SNAC with arthritis (salvage): radial styloidectomy** (stage I), **scaphoid excision + four-corner fusion (lunate, capitate, hamate, triquetrum: stages II-III)**, **proximal row carpectomy (PRC: requires intact capitate head and lunate fossa cartilage: stage I-II; contraindicated if capitolunate arthritis)**, **total wrist arthrodesis** (stage IV / pancarpal), **wrist denervation** (AIN/PIN neurectomy).

Part B: Carpal instability

6. Kinematics and ligaments

- **Proximal row = intercalated segment** (no tendon insertions): moves in response to forces. **Scaphoid tends to flex**, **triquetrum tends to extend** (helical hamate-triquetral articulation); the **lunate is balanced between them** by the **scapholunate (SL)** and **lunotriquetral (LT)** ligaments.
- **Scapholunate interosseous ligament (SLIL): dorsal part strongest (most important)**, proximal (membranous), palmar. **Secondary stabilisers: dorsal intercarpal ligament (DIC)**, **dorsal radiocarpal ligament (DRC)**, **scaphotrapeziotrapezoid (STT) ligaments**, **radioscaphocapitate (RSC)**.
- **Lunotriquetral ligament: palmar part strongest.**
- **Extrinsic palmar ligaments: radioscaphocapitate**, **long radiolunate**, **short radiolunate**, **radioscapholunate (ligament of Testut: neurovascular, not mechanical)**, **ulnolunate, ulnotriquetral**. **Space of Poirier** (**weak area between the RSC and long radiolunate, over the capitolunate joint**: **perilunate dislocations escape through it**).
- **Normal angles (lateral radiograph): scapholunate angle 30-60 degrees (mean 47)**, **radiolunate angle 0 ± 15 degrees**, **capitolunate angle 0-15 degrees**.

7. Classification of carpal instability (Mayo)

- **CID (carpal instability dissociative)**: within a row: **SL dissociation (DISI)**, **LT dissociation (VISI)**, unstable scaphoid fractures.

- **CIND (non-dissociative)**: between rows or radius-carpus: **midcarpal instability**, **radiocarpal instability** (after malunited distal radius fractures, ulnar translocation (rheumatoid)).
- **CIC (complex)**: **perilunate dislocations** (CID + CIND).
- **CIA (adaptive)**: carpal malalignment from **distal radius malunion** (dorsal tilt: adaptive DISI).

8. Scapholunate dissociation (commonest carpal instability)

- **Mechanism: FOOSH with extension, ulnar deviation and intercarpal supination**; **Mayfield stage I** of perilunate injury.
- **Signs:** dorsal SL tenderness (**just distal to Lister's tubercle**), **Watson scaphoid shift test** (**thumb pressure on the scaphoid tubercle while moving from ulnar to radial deviation: painful clunk as the scaphoid subluxes dorsally over the rim and reduces**).
- **Radiographs:**
  - **SL gap over 3 mm (Terry Thomas / David Letterman sign)** (**clenched-fist PA view** (dynamic) and comparison),
  - **Scaphoid ring (cortical ring) sign** (**flexed, foreshortened scaphoid; ring under 7 mm from the proximal pole**),
  - **SL angle over 60-70 degrees**, **DISI (dorsal intercalated segment instability: lunate extended: radiolunate angle over 15 degrees dorsally)**.
- **Investigations: MR arthrography**, **wrist arthroscopy (gold standard)**: **Geissler classification** (**I: attenuation/haemorrhage, no incongruity; II: midcarpal step-off, probe can enter; III: step-off in both radiocarpal and midcarpal joints, probe passes between bones; IV: gross instability, 2.7 mm arthroscope passes through the gap ("drive-through sign")**).
- **Garcia-Elias staging** (6-question algorithm: partial tear? repairable dorsal SL? scaphoid normally aligned? carpal malalignment reducible? cartilage normal?).
- **Treatment:**
  - **Acute (under about 6 weeks) partial (Geissler I-II): cast, arthroscopic debridement ± thermal shrinkage, percutaneous**

pinning (SL and SC K-wires).

- **Acute complete with repairable ligament: open repair (suture anchors) + K-wire stabilisation + dorsal capsulodesis (Blatt, Mayo DIC-based)**.
- **Chronic, reducible, no arthritis: reconstruction**: **3-ligament tenodesis (3LT; Garcia-Elias: modified Brunelli FCR tenodesis)**, **SLAM**, **RASL (reduction and association of the scaphoid and lunate: Herbert screw across the SL joint)**, **capsulodesis**.
- **Chronic irreducible/fixed, no arthritis: STT fusion**, **scaphocapitate fusion** (limited).
- **Arthritis: SLAC wrist**.

9. SLAC wrist (scapholunate advanced collapse) (Watson)

- **Commonest pattern of degenerative wrist arthritis.**
- **Stage I: radial styloid-scaphoid**, **Stage II: entire radioscaphoid fossa**, **Stage III: capitolunate (midcarpal) arthritis (capitate migrates proximally into the SL gap)**, **Stage IV: pancarpal (radiolunate rarely involved)**.
- **Radiolunate joint is spared** (spherical, congruent; load is perpendicular): **basis for salvage procedures that preserve it**.
- **Treatment: stage I: radial styloidectomy ± SL reconstruction/denervation**; **stage II: PRC or scaphoid excision + four-corner fusion (4CF)**; **stage III: 4CF** (PRC contraindicated: capitate head damaged); **stage IV: total wrist arthrodesis** (or arthroplasty in low-demand).
- **PRC vs 4CF: similar pain relief and grip**; **PRC: more ROM, simpler, no union needed, but more radiocapitate arthritis long-term**; **4CF: better grip/durability, longer immobilisation, non-union, dorsal impingement (lunate in DISI must be corrected)**.
- **Chondrocalcinosis (CPPD)** also causes SLAC-like changes (SLAC from CPPD; STT arthritis).

10. Lunotriquetral (LT) dissociation

- **FOOSH on the hypothenar region with wrist extension and radial deviation**; **ulnar-sided wrist pain**, **Reagan ballottement test**, **Kleinman shear test**, **Derby test**.
- **VISI (volar intercalated segment instability: lunate flexed, follows the scaphoid)** only if the dorsal radiocarpal and DIC ligaments are also injured; **break in Gilula arcs** (LT).
- **Differential: TFCC tear**, **ulnar impaction**, **ECU tendinopathy/subluxation**, **pisotriquetral arthritis**.
- **Treatment: splint, arthroscopic debridement, pinning**, **LT ligament repair/reconstruction (ECU tenodesis)**, **LT arthrodesis**, **ulnar shortening osteotomy** (tensions the ulnocarpal ligaments; especially with positive ulnar variance).

11. Midcarpal instability (CIND)

- **Palmar midcarpal instability (commonest CIND)**: **painful "clunk" with ulnar deviation and pronation** (proximal row snaps from flexion to extension); **generalised laxity**; **VISI at rest**. **Midcarpal shift test (Lichtman)**. **Treatment: splints (pisiform boost), proprioceptive training (FCU, ECU strengthening)**, **dorsal capsulodesis/ligament reconstruction**, **midcarpal (4CF) fusion** if refractory.

12. Perilunate dislocation and fracture-dislocation

- **High-energy hyperextension, ulnar deviation, intercarpal supination.**
- **Mayfield progressive perilunar instability (radial to ulnar around the lunate):**
  - **Stage I: scapholunate disruption** (or scaphoid fracture),
  - **Stage II: capitolunate dislocation** (through the **space of Poirier**),
  - **Stage III: lunotriquetral disruption** (**perilunate dislocation**: capitate dorsal to the lunate; lunate stays in the radius fossa),
  - **Stage IV: lunate dislocation** (**dorsal radiolunate ligament fails**: lunate **extruded volarly into the carpal tunnel** ("**spilled teacup sign**" on lateral), **acute median nerve compression**).

- **Greater arc injuries (through bones: trans-scaphoid, transcapitate (scaphocapitate syndrome: Fenton: capitate head rotates 180 degrees), trans-triquetral)** vs **lesser arc (pure ligamentous)**; **trans-scaphoid perilunate fracture-dislocation (de Quervain) = commonest**.
- **PA radiograph: disrupted Gilula arcs**, **triangular (instead of trapezoidal) lunate ("piece of pie" sign)** in lunate dislocation, overlapping carpal bones.
- **Missed in about 25%** initially (**review the lateral: capitate-lunate-radius colinear?**).
- **Management: urgent closed reduction (Tavernier manoeuvre)** (**median nerve symptoms = emergency**); **definitive: open reduction (dorsal ± volar approach), repair of SL/LT ligaments, fixation of fractures (scaphoid screw), K-wires (SL, LT, scaphocapitate), carpal tunnel release (volar approach)**. **Outcomes:** post-traumatic arthritis (**about 50%+ radiographically**), stiffness, but acceptable function.
- **Chronic (over 6-8 weeks) missed perilunate dislocations:** open reduction possible (± shortening); **PRC** or **wrist arthrodesis** as salvage.

13. Kienböck disease (lunate AVN) (brief)

- **Young men (20-40)**, **negative ulnar variance (association)**, **single vessel supply (Y/I pattern)**, repetitive loading.
- **Lichtman staging: I normal radiograph (MRI changes)**, **II lunate sclerosis**, **IIIA collapse without carpal collapse (normal radioscaphoid angle)**, **IIIB collapse with fixed scaphoid flexion (ring sign; radioscaphoid angle over 60 degrees)**, **IIIC coronal fracture of the lunate**, **IV pancarpal arthritis**.
- **Treatment: stage I: immobilisation**; **II-IIIA: joint levelling (radial shortening osteotomy with negative ulnar variance; capitate shortening or radial wedge osteotomy with neutral/positive variance)**, **revascularisation (4,5-ECA pedicled vascularised graft)**; **IIIB: STT or scaphocapitate fusion, PRC**; **IV: wrist arthrodesis, PRC (if capitate head intact)**. **Children: non-operative (good prognosis)**.
- **Preiser disease** = idiopathic **scaphoid AVN**.

Exam pearl: **Scaphoid: dorsal carpal branch of the radial artery enters at the dorsal ridge, retrograde to the proximal pole (AVN). Occult fracture: MRI. Displacement over 1 mm, intrascaphoid angle over 35, proximal pole = fix (headless screw, central axis; volar for waist, dorsal for proximal pole). SWIFFT: cast undisplaced waist fractures, fix non-unions.**

Exam pearl: **Non-union with AVN (no punctate bleeding): vascularised graft (1,2-ICSRA; MFC free graft with humpback). SNAC: styloid-scaphoid, then scaphocapitate, then capitolunate; radiolunate spared. SLAC: same sparing. PRC needs intact capitate head; stage III = 4CF.**

Exam pearl: **SL dissociation: Watson shift test, SL gap over 3 mm, ring sign, SL angle over 60-70 degrees, DISI; Geissler IV = drive-through. Mayfield I SL, II capitolunate (space of Poirier), III LT (perilunate), IV lunate dislocation (spilled teacup, median nerve). Greater arc through bones (trans-scaphoid commonest).**

Hand infections

1. General principles

- **Commonest organism: *Staphylococcus aureus*** (**MRSA increasingly common in community-acquired hand infections**), then **beta-haemolytic streptococci**; **mixed/anaerobic in bites, diabetics, IV drug users**.
- **Hand anatomy predisposes to spread:** closed compartments (pulp, sheaths, deep palmar spaces), **lymphatic drainage to the dorsum (dorsal swelling even with palmar infection)**.
- **Principles: elevation, immobilisation in the "safe position" (intrinsic plus: wrist extended about 30 degrees, MCPJs flexed 70-90 degrees, IPJs extended)**, **antibiotics**, **early surgical drainage of pus**, **debridement**, **early mobilisation** once infection controlled; **check tetanus status**; **control diabetes**.
- **Abscess (fluctuant collection) = drainage; cellulitis = antibiotics + elevation** (**reassess within 24-48 hours; failure to improve: suspect abscess**).

2. Paronychia

- **Infection of the nail fold (perionychium)**: **commonest hand infection**; **acute: *S. aureus*** (nail biting, manicure, hangnail).
- **Treatment: early (cellulitis): warm soaks, oral antibiotics**; **abscess: incision/elevation of the nail fold (blade directed away from the nail bed), partial nail plate removal if pus beneath the nail (subungual abscess)**.
- **Eponychia** (proximal nail fold), **"run-around" infection** (involves both lateral folds and the eponychium).
- **Chronic paronychia: Candida albicans** (**wet work: dishwashers, bartenders, nurses, diabetics**), **inflammatory**; **avoid moisture, topical antifungals/steroids**, **eponychial marsupialisation** (crescent excision of the dorsal eponychial fold) **± nail removal** for refractory cases.

- **Differentials: herpetic whitlow**, **subungual melanoma** (persistent "paronychia"/pigment: biopsy), **squamous cell carcinoma**, **pyogenic granuloma**.

3. Felon (pulp space abscess)

- **Closed-space infection of the distal pulp**, divided into **compartments by fibrous septa** (from the skin to the periosteum of the distal phalanx): **pressure: necrosis, osteomyelitis of the distal phalanx, septic DIPJ, flexor sheath infection**.
- **Organism: *S. aureus***; **penetrating injury** (finger-prick glucose testing).
- **Throbbing pain, tense swollen pulp**.
- **Treatment: incision and drainage** with a **high lateral (mid-axial) incision** on the **non-contact side** (**ulnar side of index/middle/ring; radial side of thumb and little finger**), dividing septa, **avoiding the neurovascular bundles and the flexor sheath (stay distal to the DIP crease)**, or a **volar longitudinal incision** over the point of maximum fluctuance. **Avoid "fish-mouth" (painful scar, unstable pulp) and through-and-through incisions.**

4. Herpetic whitlow

- **Herpes simplex virus (HSV-1 or HSV-2)** infection of the fingertip; **healthcare workers (dentists, respiratory therapists), children (thumb suckers with gingivostomatitis)**.
- **Intense pain, clear vesicles (clustered), on an erythematous base**, **non-tense pulp**, prodrome of burning/tingling; **vesicles coalesce, become cloudy**.
- **Diagnosis:** clinical; **Tzanck smear (multinucleated giant cells)**, **viral PCR/culture**.
- **Treatment: non-operative**: **self-limiting (about 3 weeks)**, **protective dressing**, **oral acyclovir (if within 48-72 hours)**; **DO NOT incise** (**risk of secondary bacterial infection, systemic viraemia (encephalitis)**).

5. Pyogenic flexor tenosynovitis

- **Infection within the synovial flexor tendon sheath**: **surgical emergency** (**tendon necrosis, adhesions, rupture**, spread to deep spaces).
- **Cause: penetrating injury (commonest), haematogenous spread (*Neisseria gonorrhoeae*: disseminated gonococcal infection in sexually active young adults)**.
- **Organisms: *S. aureus*** (commonest), streptococci, mixed (**diabetics, bites**), **gonococcus**, *Mycobacterium marinum* (**fish tanks: subacute chronic tenosynovitis**: granulomas; culture at **30-32 C**; **clarithromycin/ethambutol, rifampicin**).
- **Kanavel's four cardinal signs:**
  - **Finger held in slight (semi-)flexion**,
  - **Fusiform (symmetrical) swelling** of the entire digit ("sausage digit"),
  - **Tenderness along the course of the flexor sheath**,
  - **Pain on passive extension of the finger** (**earliest and most sensitive sign**; along the sheath).
- **Sheath anatomy and spread:**
  - **Index, middle and ring finger sheaths end at the A1 pulley (MCPJ level)**.
  - **Thumb sheath (FPL) is continuous with the radial bursa**; **little finger sheath is continuous with the ulnar bursa**; the radial and ulnar bursae **communicate in about 50-80%** (**via Parona's space**: between the pronator quadratus and FDP in the distal forearm): **"horseshoe abscess"** (thumb and little finger infection together).
- **Treatment:**
  - **Early (under about 24-48 hours) with equivocal signs**: **IV antibiotics, splinting, elevation**: **reassess within 12-24 hours**; **surgery if no improvement**.
  - **Established infection: surgical drainage**: **closed catheter irrigation (Neviaser)**: **proximal incision at the A1 pulley (palmar) and distal incision at the A5 pulley (DIP crease / mid-axial)**, **catheter irrigation of the sheath** (**limited open drainage**); **open debridement**

(Bruner/mid-axial incisions) for severe/late infections with necrotic tendon.
  - **Post-op:** IV antibiotics, early ROM, elevation.
- **Complications: stiffness/adhesions (commonest)**, tendon necrosis/rupture, **amputation (diabetes, peripheral vascular disease, renal failure, subcutaneous purulence, delayed presentation)**: **Pang classification** of outcomes.

6. Deep space infections of the hand

Thenar space

- **Boundaries: radial to the oblique septum (from the third metacarpal) between the flexor tendons of the index and the adductor pollicis**; **lies deep to the index flexor tendons, superficial to the adductor pollicis**.
- **Signs: thumb held abducted** (**widened first web space**), **swelling of the thenar eminence and first web**, **pain on thumb opposition/flexion**.
- **Drainage: dual incisions: volar (thenar crease parallel, avoid the recurrent motor branch) and dorsal (first web, longitudinal)**; **avoid incisions across the first web (contracture)**.

Midpalmar space

- **Ulnar to the oblique septum**, **deep to the flexor tendons of the middle, ring and little fingers**, superficial to the interossei/metacarpals.
- **Signs: loss of palmar concavity** (convexity), **dorsal hand swelling**, painful middle and ring finger motion.
- **Drainage: palmar transverse/curved incision**.

Hypothenar space

- **Rare**; **ulnar border of the palm**; incision on the ulnar side.

Collar-button (web space) abscess

- **Infection of the interdigital web space** (**fissure or callus in labourers**): **dumbbell shape (volar and dorsal components connected through the palmar fascia)**: **abducted adjacent fingers**, **dorsal swelling**.

- **Drainage: volar (zig-zag or longitudinal) and dorsal (longitudinal) incisions**; **do not incise across the web (contracture)**.

Parona's space infection

- **Distal forearm, between pronator quadratus and FDP**: from **radial/ulnar bursa spread**; **forearm swelling, pain with finger flexion**, **acute carpal tunnel syndrome**; drainage with **carpal tunnel release**.

7. Septic arthritis of the hand and wrist

- **Penetrating trauma, bites (MCPJ: fight bite), haematogenous**; **pain on axial loading and motion**, joint swelling.
- **Aspirate** (diagnostic; small joints may be dry), **radiograph** (foreign body, bony involvement).
- **Treatment: surgical irrigation and debridement (arthrotomy/arthroscopy for wrist)**, **IV antibiotics**; **wrist septic arthritis: consider *Neisseria gonorrhoeae* (young sexually active), crystal disease (gout/CPPD: differential)**.

8. Bites

- **Human bites / fight bite (clenched-fist injury)**: see extensor tendons: **MCPJ**, ***Eikenella corrodens***, **streptococci, *S. aureus*, anaerobes (Prevotella, Fusobacterium)**; **co-amoxiclav**; **explore, irrigate, leave open**; **do not close primarily**.
- **Dog bites: *Pasteurella multocida*** (**also cats**), ***Capnocytophaga canimorsus*** (**asplenic, alcoholics: fulminant sepsis**), *S. aureus*, streptococci, anaerobes; **crush injury** component.
- **Cat bites: *Pasteurella multocida* (rapid onset, within 12-24 hours)**: **deep puncture wounds (sharp teeth) penetrate joints/sheaths: higher infection rate (about 30-50%) than dog bites**; **cat scratch disease** (***Bartonella henselae***: regional lymphadenopathy).
- **Antibiotic of choice for animal bites: amoxicillin-clavulanate (co-amoxiclav)**; **penicillin allergy: doxycycline (or ciprofloxacin/co-trimoxazole) + clindamycin/metronidazole**. ***Pasteurella* is resistant to**

first-generation cephalosporins (cefalexin), flucloxacillin, clindamycin and erythromycin (alone).

- **Rabies prophylaxis** (dogs, bats, monkeys; India: high burden: **post-exposure prophylaxis: wound washing, rabies vaccine + rabies immunoglobulin (category III)**), **tetanus**.
- **Rat bites**: ***Streptobacillus moniliformis*** (**rat-bite fever**), ***Spirillum minus***.
- **Snake bite** (India: **Big Four: Indian cobra, common krait, Russell's viper, saw-scaled viper**): local tissue necrosis; **compartment syndrome is rare: fasciotomy only with measured raised pressures** (not routinely); **anti-snake venom (ASV)** (polyvalent).

9. Necrotising fasciitis

- **Rapidly spreading infection of the fascia and subcutaneous tissue**: **surgical emergency**; **mortality about 20-30%**.
- **Types: Type I (polymicrobial: commonest; diabetics, IV drug users)**; **Type II (monomicrobial: group A *Streptococcus pyogenes* ± *S. aureus*/MRSA)**: **toxic shock**; **Type III (*Vibrio vulnificus*: seawater, seafood; Clostridium: gas gangrene)**; (Type IV fungal).
- **Signs: pain out of proportion to clinical findings (earliest)**, **rapid spread**, **tense oedema beyond erythema**, **skin bullae, crepitus, grey/dusky skin, anaesthesia**, **systemic toxicity (sepsis)**.
- **LRINEC score (Laboratory Risk Indicator for Necrotising Fasciitis)**: **CRP, WBC, haemoglobin, sodium, creatinine, glucose**: **score ≥6 suspicious; ≥8 high risk** (**low sensitivity: do not rely on it to exclude**).
- **"Finger test"** at the bedside/operation: **lack of bleeding, "dishwater" fluid, fascia separates easily from tissue with finger dissection**.
- **Treatment: emergency radical surgical debridement (repeat every 24-48 hours)**, **broad-spectrum IV antibiotics (carbapenem/piperacillin-tazobactam + clindamycin (toxin suppression) + vancomycin/linezolid for MRSA)**, **ICU support**, **IV immunoglobulin (streptococcal toxic shock)**, **amputation** if needed; **hyperbaric oxygen** (adjunct, unproven).

10. Atypical and chronic infections

- ***Mycobacterium marinum*** (**"fish tank granuloma"**: aquarium workers, fishermen): **chronic tenosynovitis, nodules along lymphatics (sporotrichoid spread)**; **culture at 30-32 C (Lowenstein-Jensen)**, **biopsy (granulomas)**; **clarithromycin + ethambutol (± rifampicin) for 3-6 months**, synovectomy.
- ***M. tuberculosis***: **compound palmar ganglion (TB tenosynovitis of the flexor sheaths and radial/ulnar bursae: "rice bodies", hourglass swelling above and below the flexor retinaculum: cross-fluctuation)**; **tuberculous dactylitis (spina ventosa: children; expansile phalangeal/metacarpal lesion)**; **anti-tubercular therapy (ATT) + synovectomy**.
- **Sporotrichosis** (***Sporothrix schenckii***: **rose gardeners' disease, thorn pricks**): **nodular lymphangitis**; **itraconazole**.
- **Atypical mycobacteria** (*M. avium-intracellulare*, *M. kansasii*): immunocompromised.
- **Fungal: onychomycosis (*Trichophyton rubrum*)**: oral terbinafine.
- **Leprosy (*Mycobacterium leprae*)**: **ulnar nerve (commonest), median nerve (claw hand)**, **thickened nerves**, **neuropathic hands**; tendon transfers (**Indian experience: Brand, Fritschi**) after the disease is inactive.
- **Orf (parapoxvirus: sheep/goats)**, **milker's nodules**: self-limiting.
- **Anthrax** (rare; black eschar).
- **Gonococcal tenosynovitis/septic arthritis**: **migratory polyarthralgia, tenosynovitis (dorsal hand/wrist), pustular skin lesions** (disseminated gonococcal infection); **ceftriaxone**.

11. Special hosts

- **Diabetics:** polymicrobial (**gram-negatives, anaerobes**), atypical presentations, **higher amputation risk**, **worse outcomes in flexor tenosynovitis**; **tight glucose control**.
- **IV drug users: MRSA**, **Eikenella** (licking needles), **pseudomonas**, **necrotising fasciitis**, **injection into the dorsal hand/forearm**; **high-**

pressure injection injuries (see below).

- **Immunocompromised/HIV**: atypical organisms, fungi.

12. High-pressure injection injuries

- **Paint, grease, hydraulic fluid, solvents** under high pressure (**over 7,000 kPa / about 1,000 psi**): **index finger of the non-dominant hand** (commonest).
- **Innocuous-looking pinhole entry wound** with **extensive tracking along tissue planes and sheaths**; **chemical inflammation, ischaemia, secondary infection**.
- **Surgical emergency: wide surgical debridement (decompression) within 6 hours**; **oil-based paints/solvents cause the most damage (higher amputation rates up to about 50-60%)**; **water and air** less damaging; **grease**: intermediate.
- **Amputation risk** increases with **delay (over 6-10 hours)**, **oil-based paint**, **higher pressure**, **injection into the digit (vs palm)**.
- **Radiographs** (lead-based paint visible; air), **no role for steroids (uncertain)**, broad-spectrum antibiotics.

Exam pearl: **Kanavel signs: flexed posture, fusiform swelling, sheath tenderness, pain on passive extension (earliest/most sensitive). Thumb sheath = radial bursa; little finger = ulnar bursa; communicate via Parona's space (horseshoe abscess). Closed catheter irrigation (A1 to A5).**

Exam pearl: **Felon: mid-axial incision on the non-contact side, divide septa; avoid fish-mouth. Herpetic whitlow: do not incise. Chronic paronychia: Candida (wet work); marsupialisation. Thenar space: thumb abducted; midpalmar: loss of palmar concavity; collar-button: dual incisions, never across the web.**

Exam pearl: **Bites: human = Eikenella; dog/cat = Pasteurella (cat bites infect more); Capnocytophaga in asplenics; all = co-amoxiclav. Necrotising fasciitis: pain out of proportion, LRINEC ≥6 (insensitive), finger test, radical debridement + clindamycin. M. marinum: fish tank, culture at 30-32 C. Compound palmar ganglion = TB with rice bodies. High-pressure injection: oil-based paint worst; emergency debridement.**

Dupuytren and rheumatoid hand

Part A: Dupuytren disease

1. Epidemiology and associations

- **Benign fibroproliferative disorder of the palmar (and digital) fascia** leading to **nodules, cords and flexion contractures**.
- **Northern European (Celtic/Viking) ancestry**, **men (about 6-10:1)**, **older age (over 50)**, **autosomal dominant with variable penetrance**; **less common in Asians and Africans** (but seen in India).
- **Associations: alcohol (liver disease)**, **smoking**, **diabetes** (**milder: ring and middle fingers, palmar nodules**), **epilepsy (anticonvulsants: phenytoin, phenobarbital)**, **HIV**, **manual labour/vibration (debated)**, **hyperlipidaemia**, **previous hand trauma**.
- **Dupuytren diathesis (Hueston) (aggressive disease, high recurrence):**
  - **Early onset (under 50)**,
  - **Family history (positive)**,
  - **Bilateral disease**,
  - **Ectopic lesions: Garrod's knuckle pads (dorsal PIPJ)**, **Ledderhose disease (plantar fibromatosis)**, **Peyronie disease (penile fibromatosis)**,
  - **Male sex, Northern European ethnicity, radial-sided disease**.

2. Pathology

- **Myofibroblast** (**alpha-smooth muscle actin**: contractile) is the key cell.
- **Increased type III collagen (relative to type I)** (normal fascia: mainly type I).
- **Growth factors: TGF-beta (key), PDGF, basic FGF**; **Wnt signalling** genetic loci; **TNF** (anti-TNF trials for early nodules).
- **Luck stages:**
  - **Proliferative** (myofibroblasts; nodules; highly cellular),
  - **Involutional** (myofibroblasts align along tension lines; cords form),

- **Residual** (acellular, tendon-like cords; fibrocytes, type I collagen).

3. Normal fascia becomes diseased cords

Normal structure	Diseased cord	Effect
Pretendinous band	**Pretendinous cord**	**MCPJ flexion contracture**
Spiral band + **lateral digital sheet** + **Grayson ligament** + pretendinous band	**Spiral cord**	**PIPJ contracture**; **displaces the neurovascular bundle centrally (midline), superficially and proximally** (**at risk during surgery**)
Lateral digital sheet	**Lateral cord**	**PIPJ / DIPJ contracture**
Natatory ligament	**Natatory cord**	**Web space contracture (adduction)**
Central cord (no normal precursor: from the pretendinous band continuation)	**Central cord**	**PIPJ contracture**
Retrovascular band (Thomine)	**Retrovascular cord**	**DIPJ contracture** (residual after fasciectomy)
Abductor digiti minimi cord (little finger)	**Abductor digiti minimi cord**	**PIPJ contracture** of little finger
First web: proximal commissural ligament	**Commissural cords**	**First web contracture**

- **Cleland ligaments (dorsal to the neurovascular bundle) are NOT involved** in Dupuytren disease ("**Cleland's never Clings**"); **Grayson ligaments (volar to the bundle) ARE involved**.
- **Transverse fibres of the palmar aponeurosis: not involved** (usually).
- **Spiral cord: a spiral nerve should be suspected with a soft-tissue mass between the MCP and PIP flexion creases, a PIPJ contracture, and interdigital soft-tissue fullness.**
- **Fingers involved: ring (commonest) > little > middle > thumb > index**.

4. Clinical evaluation

- **Painless nodules in the palm (distal palmar crease, ring finger)**, **cords**, **skin pitting**, **flexion contractures (MCPJ, PIPJ)**.
- **Hueston tabletop test**: **inability to place the hand flat on a table = indication for surgery** (positive test).
- **Measure joint contractures with a goniometer.**
- **Tubiana classification (total flexion deformity per ray): stage 0** no lesion; **N** nodule without contracture; **stage 1: 0-45 degrees**, **2: 45-90**, **3: 90-135**, **4: over 135 degrees**.

5. Indications for intervention

- **MCPJ contracture of 30 degrees or more** (**MCPJ contractures correct well because the collateral ligaments are tight in flexion: no capsular contracture**).
- **Any PIPJ contracture (of about 15-20 degrees or more)** (**PIPJ contractures are harder to correct and recur: secondary capsular/volar plate/collateral ligament contracture, central slip attenuation**; **intervene earlier**).
- **Functional impairment, positive tabletop test.**
- **Nodules alone: not an indication** (unless painful).

6. Treatment options

Non-surgical

- **Splinting alone: ineffective**; **steroid injection** into painful nodules.
- **Radiotherapy** (**early nodular disease**: may slow progression).

Percutaneous needle fasciotomy (PNF/needle aponeurotomy)

- **Office-based**, local anaesthetic, divide the cord with a needle bevel at multiple points; **best for MCPJ contractures with well-defined palmar cords**.
- **Fast recovery, low complication rate**, **but recurrence higher (about 50-85% at 5 years)**; **repeatable**; **nerve injury and tendon injury risks** (keep the needle superficial; patient feedback).

Collagenase *Clostridium histolyticum* (CCH) injection

- **Injection into the cord (enzymatic lysis: type I and III collagen)**, **manipulation 24-72 hours later** to rupture the cord.
- **Results: good correction of MCPJ (about 75-80% to under 5 degrees), less for PIPJ**; **recurrence about 35-50% at 5 years**.
- **Complications: skin tears (commonest significant: during manipulation)**, **oedema, bruising, lymphadenopathy, pain**, **flexor tendon rupture (rare: ring/little finger PIPJ injections)**, **pulley rupture**, **CRPS**, **allergic reactions**.
- (Commercial availability has varied by country.)

Surgery

- **Limited (regional/selective) fasciectomy** (**the standard operation: excise the diseased fascia only**): **Bruner zig-zag incision or longitudinal incision with Z-plasties**; **recurrence about 20-40% at 5 years**; **PIPJ correction is less reliable**.
- **Segmental fasciectomy** (**excision of short segments of the cord through small incisions**).
- **Dermofasciectomy** (**excision of diseased fascia + overlying skin, replaced with a full-thickness skin graft**): **recurrent disease, diathesis, young patients with aggressive disease, skin involvement**; **lowest recurrence (under graft: "firebreak")**.
- **Open palm technique (McCash)**: palmar transverse incision left open to heal by secondary intention (**reduces haematoma, swelling and pain; useful in elderly**).
- **Additional PIPJ procedures** for residual contracture: **check-rein ligament release, volar plate release, accessory collateral ligament release**; **beware of central slip attenuation (residual extension lag)**.
- **Salvage: PIPJ arthrodesis**, **amputation** (multiple recurrences, severe little finger contractures), **dynamic external fixation (Digit Widget/TEC: continuous elongation: severe PIPJ contractures)**.
- **Post-operative: night extension splint (common, evidence of benefit limited)**, hand therapy.

Comparisons

- **Recurrence: dermofasciectomy < limited fasciectomy < collagenase ≈ needle fasciotomy.**
- **Recovery and complications: needle/collagenase faster but higher recurrence**; **surgery more durable but longer recovery**. **DISC trial (UK)**: collagenase vs limited fasciectomy: **surgery slightly better** clinically.

7. Complications of surgery

- **Digital nerve injury (about 2-3% primary; up to about 10% in revision)** (spiral cord), **digital artery injury (devascularisation: white finger after release: release tension (flex the finger), warm, topical vasodilators; microvascular repair)**, **haematoma**, **skin necrosis (flap tips)**, **infection**, **"flare reaction" (pain, swelling, stiffness: pseudo-CRPS: women)**, **CRPS (about 5%)**, **recurrence (true recurrence in the operated field) vs extension (new disease outside the field)**, **loss of flexion**, **PIPJ extension lag**.

Part B: Rheumatoid hand and wrist

8. General principles

- **Chronic symmetrical inflammatory polyarthritis**; **synovitis**: **pannus (proliferating synovium)** erodes cartilage, bone, **ligaments and tendons**; **tendon rupture by attrition or direct infiltration**.
- **Modern DMARDs/biologics** have **reduced the need for surgery**; **surgery: relieve pain, prevent tendon rupture (synovectomy), restore function, correct deformity**.
- **Priority of surgery (general principles): lower limb before upper limb**, **proximal before distal** (shoulder/elbow before wrist/hand), **wrist before fingers (stable wrist = foundation)**, **painful procedures first**; **joints before tendons** (or tendon transfers after joint stabilisation); **prophylactic tenosynovectomy** to prevent rupture (**over 4-6 months of persistent tenosynovitis despite medical therapy**).
- **Cervical spine assessment (atlantoaxial instability) before anaesthesia (flexion-extension radiographs)**; **perioperative DMARD**

management (methotrexate continued; biologics withheld around surgery).

9. Wrist

- **Caput ulnae syndrome**: **dorsal subluxation of the ulnar head** (**DRUJ synovitis with destruction of the TFCC and ECU subsheath; ECU subluxates volarly**: becomes a wrist flexor), **supination of the carpus** off the ulna, **radial deviation of the wrist** (**contributes to ulnar deviation of the fingers: "zig-zag deformity"**), **"piano key sign"** (ulnar head ballottement).
- **Carpal supination, volar subluxation, ulnar translocation of the carpus** (radiocarpal ligament destruction).
- **Vaughan-Jackson syndrome** (**attrition rupture of the ulnar extensors (EDM first, then EDC little and ring) over the prominent dorsal ulnar head**).
- **Mannerfelt-Norman syndrome** (**attrition rupture of the FPL** (then FDP index) **over a scaphoid/STT osteophyte** in the carpal tunnel).
- **Differentials of loss of finger extension in RA:**
  - **Extensor tendon rupture** (**tenodesis test: passive wrist flexion fails to extend the MCPJs**),
  - **PIN palsy** (**elbow synovitis compressing the PIN**; **tenodesis effect intact**),
  - **Extensor tendon subluxation into the ulnar gutter at the MCPJ** (**sagittal band attenuation: patient can maintain extension once passively extended**),
  - **MCPJ volar dislocation/subluxation** (cannot be passively extended).
- **Surgical options:**
  - **Dorsal tenosynovectomy + extensor retinaculum transposed beneath the tendons** (protects them).
  - **Distal ulna procedures: Darrach (distal ulna excision)** (**low-demand elderly; risk of ulnar stump instability/convergence**), **Sauvé-Kapandji (DRUJ arthrodesis + proximal ulnar pseudarthrosis)** (**younger; preserves ulnar support for the carpus: prevents ulnar translocation**), **ulnar head replacement**.

- **Tendon transfer/reconstruction for ruptures: single EDC rupture: side-to-side transfer to adjacent EDC**; **two tendons (little and ring): side-to-side + EIP transfer**; **three or more: FDS transfer (ring/middle) or free grafts**; **EPL rupture: EIP to EPL**; **FPL rupture: FDS ring transfer or IPJ fusion**.
- **Radiolunate (Chamay) fusion** (**prevents ulnar translocation; partial fusion when the midcarpal joint is preserved**).
- **Total wrist arthrodesis** (**gold standard for the destroyed rheumatoid wrist**; **Mannerfelt pin, Steinmann pin or plate**; **fuse in neutral or slight extension; if bilateral, consider one arthroplasty or fuse one in slight flexion for perineal hygiene**).
- **Total wrist arthroplasty**: **low-demand, bilateral disease, good bone stock**; preserves motion; higher complication rates.
- **Larsen grading** of radiological destruction (0-5).

10. MCP joints

- **Ulnar drift and volar subluxation** (**the commonest finger deformity in RA**):
  - **Synovitis stretches the radial sagittal band and collateral ligaments**,
  - **Extensor tendons sublux ulnarly into the valleys between the metacarpal heads**,
  - **Radial deviation of the wrist (with metacarpals) + ulnar pull of flexor tendons (enter the finger obliquely) + intrinsic contracture (ulnar intrinsics)** + **gravity/pinch forces** (**thumb pushes the index ulnarly**),
  - **Volar plate and accessory collateral ligament destruction**: **volar subluxation of the proximal phalanx**.
- **Treatment:**
  - **Early (no joint destruction): synovectomy, extensor tendon centralisation (sagittal band reefing), intrinsic release (cross-intrinsic transfer: ulnar intrinsic of one finger to the radial lateral band of the adjacent finger)**.

- **Joint destruction: silicone MCPJ arthroplasty (Swanson: a flexible spacer; encapsulation)** with **soft-tissue rebalancing (radial collateral ligament reconstruction, centralisation, intrinsic release)**: **pain relief, improved arc and appearance (arc about 40-60 degrees, shifted into extension)**; **implant fracture (about 10-30% at 10 years), silicone synovitis (rare at the MCPJ), recurrent ulnar drift**. **Correct wrist radial deviation first** (otherwise recurrence).
- **Pyrocarbon/surface replacement**: requires intact soft tissues (less used in RA).

11. Finger deformities

Swan-neck deformity (PIPJ hyperextension + DIPJ flexion)

- **Origins: MCPJ (volar subluxation/intrinsic tightness: commonest in RA)**, **PIPJ (volar plate laxity, synovitis, FDS rupture)**, **DIPJ (mallet: terminal tendon rupture)**.
- **Nalebuff classification (RA swan-neck):**
  - **Type I: PIPJ flexible in all MCPJ positions**: **splint (figure-of-eight/ring splint)**, **DIPJ fusion (mallet)**, **dermodesis, flexor tenodesis (FDS sling), SORL (spiral oblique retinacular ligament reconstruction)**,
  - **Type II: PIPJ flexion limited with the MCPJ extended (intrinsic tightness: positive Bunnell-Littler/Finochietto test)**: **intrinsic release** + type I procedures,
  - **Type III: PIPJ stiff in all positions, joint surface preserved**: **PIPJ manipulation, lateral band mobilisation, skin release (dorsal)**, **flexor tenosynovectomy**,
  - **Type IV: PIPJ stiff with joint destruction**: **PIPJ arthrodesis (index/middle: lateral pinch stability) or arthroplasty (ring/little)**.

Boutonniere deformity (PIPJ flexion + DIPJ hyperextension)

- **PIPJ synovitis**: **central slip attenuation**, **lateral bands migrate volarly** (triangular ligament stretched).
- **Nalebuff/Millender stages:**
  - **Stage I (mild: 10-15 degree PIPJ extension lag, passively correctable)**: **splinting**, **synovectomy**, **Fowler/Dolphin terminal**

tendon tenotomy (DIPJ hyperextension),

- **Stage II (moderate: about 30-40 degree lag, passively correctable)**: **central slip reconstruction (Matev/Curtis), lateral band relocation**,
- **Stage III (fixed, joint destroyed)**: **PIPJ arthrodesis (index/middle) or arthroplasty**.

Thumb deformities (Nalebuff classification: types I-VI)

- **Type I: boutonniere thumb (commonest: about 50-70%)**: **MCPJ flexion + IPJ hyperextension** (**MCPJ synovitis stretches the EPB insertion and dorsal capsule; EPL subluxes ulnarly/volarly**): **early: synovectomy + EPL rerouting (to the dorsal base of the proximal phalanx)**; **late: MCPJ arthrodesis (± IPJ arthrodesis/arthroplasty)**.
- **Type II: rare (boutonniere + CMCJ subluxation)**.
- **Type III: swan-neck thumb (second commonest)**: **CMCJ subluxation/adduction contracture**: **MCPJ hyperextension + IPJ flexion** (like OA): **CMCJ arthroplasty (trapeziectomy) + MCPJ fusion/volar capsulodesis**, **first web release**.
- **Type IV: gamekeeper's (UCL laxity: MCPJ radial deviation, secondary adduction)**: **UCL reconstruction or MCPJ arthrodesis**.
- **Type V: MCPJ volar plate laxity (MCPJ hyperextension, IPJ flexion; CMCJ normal)**: **volar capsulodesis/MCPJ fusion**.
- **Type VI: arthritis mutilans (opera-glass hand)**: **bone resorption, shortening, "telescoping"**: **arthrodesis with bone grafting** to restore length.

12. Other inflammatory hand conditions

- **Psoriatic arthritis: DIPJ involvement**, **pencil-in-cup deformity**, **arthritis mutilans**, **dactylitis (sausage digit)**, **nail pitting**, **asymmetric**; **stiffness (ankylosis) rather than instability**; **avoid surgery through active psoriatic plaques (infection risk)**; **arthrodesis in functional position** commonly.
- **Systemic lupus erythematosus (SLE): ligamentous laxity with reducible deformities (Jaccoud arthropathy: ulnar drift, swan neck)**

WITHOUT joint destruction: **soft-tissue procedures fail (recurrence): arthrodesis often preferred**.

- **Scleroderma (systemic sclerosis): PIPJ flexion contractures** (**dorsal skin ulceration over the PIPJ**), **Raynaud phenomenon, digital ischaemic ulcers**, **calcinosis (CREST)**, **acro-osteolysis**; **PIPJ arthrodesis (with shortening)**, **digital sympathectomy (periarterial)**, **botulinum toxin for Raynaud**, **calcinosis excision**.
- **Gout: tophi (urate deposits)**, **flexor tenosynovitis, carpal tunnel syndrome**, **punched-out erosions with overhanging edges ("rat bite")**; **debulking of tophi** for skin ulceration, infection, nerve compression.
- **Osteoarthritis of the hand (comparison): Heberden nodes (DIPJ)**, **Bouchard nodes (PIPJ)**, **thumb CMCJ (basal joint) OA (Eaton-Littler stages I-IV: trapeziectomy ± LRTI for stages II-IV; CMCJ arthrodesis in young manual workers)**, **mucous cyst (DIPJ OA ganglion: excise with osteophyte)**.

Exam pearl: **Dupuytren: myofibroblast, type III collagen up, TGF-beta. Diathesis: early onset, family history, bilateral, ectopic (Garrod, Ledderhose, Peyronie). Cleland ligament spared; Grayson involved. Spiral cord (pretendinous band + spiral band + lateral digital sheet + Grayson) displaces the NV bundle central, superficial, proximal.**

Exam pearl: **Operate for MCPJ over 30 degrees, any PIPJ contracture, positive tabletop test. Recurrence: dermofasciectomy lowest; needle and collagenase highest. Collagenase: skin tears; tendon rupture rare. White finger after release: flex finger, warm.**

Exam pearl: **RA: caput ulnae (dorsal ulnar head, ECU volar), Vaughan-Jackson (EDM first), Mannerfelt (FPL over scaphoid). Loss of finger extension: rupture (tenodesis test negative), PIN palsy (tenodesis intact), subluxed tendon (holds once placed), MCPJ dislocation. Darrach (elderly) vs Sauvé-Kapandji (young). Wrist fusion is gold standard; correct wrist radial deviation before MCPJ silicone arthroplasty.**

Exam pearl: **Swan-neck Nalebuff I flexible, II intrinsic tight, III stiff/joint preserved, IV destroyed. Boutonniere thumb (type I) commonest RA thumb deformity. SLE: reducible deformities without destruction (Jaccoud); soft-tissue procedures fail. Psoriatic: DIPJ, pencil-in-cup. Scleroderma: PIPJ flexion, calcinosis.**

Congenital hand

1. Embryology

- **Upper limb bud appears at about day 26-28 of gestation**; **limb development is complete by about week 8** (most congenital differences arise between **weeks 4 and 8**).
- **Signalling centres (three axes):**

Axis	Signalling centre	Molecule	Defects
Proximal-distal	**Apical ectodermal ridge (AER)**	**Fibroblast growth factors (FGF-4, FGF-8)**	**Transverse deficiencies (amputations)**, **symbrachydactyly**
Anterior-posterior (radial-ulnar)	**Zone of polarising activity (ZPA)** (posterior mesoderm)	**Sonic hedgehog (SHH)**	**Mirror hand (ulnar dimelia: SHH duplication)**, **polydactyly**, **radial/ulnar deficiencies**
Dorsal-ventral	**Dorsal ectoderm** (**Wnt-7a**) / ventral ectoderm (**engrailed-1 (EN-1)** suppresses Wnt-7a)	**Wnt-7a**, **LMX1B** (dorsal mesoderm)	**Nail-patella syndrome (LMX1B: absent/hypoplastic nails, absent patella)**, **dorsal dimelia (palmar nails)**

- **Digit separation: apoptosis** of interdigital tissue (**bone morphogenetic proteins (BMPs)**) at about **weeks 6-8**: failure = **syndactyly**.
- **HOX genes** (HOXA13, HOXD13: **synpolydactyly**).

2. Classification

- **Swanson (IFSSH, 1976) classification (historical, 7 groups):**
  - **Failure of formation** (transverse: congenital amputations; longitudinal: radial/ulnar/central deficiencies),
  - **Failure of differentiation (separation)** (syndactyly, camptodactyly, clinodactyly, arthrogryposis, synostosis),
  - **Duplication** (polydactyly, mirror hand),

  - **Overgrowth** (macrodactyly),
  - **Undergrowth** (thumb hypoplasia, brachydactyly),
  - **Constriction band syndrome**,
  - **Generalised skeletal abnormalities**.
- **OMT classification (Oberg-Manske-Tonkin; adopted by IFSSH 2014)**: **Malformations** (abnormal axis formation/differentiation: **proximal-distal, radial-ulnar, dorsal-ventral, unspecified axis**), **Deformations** (constriction ring), **Dysplasias** (hypertrophy: macrodactyly; tumorous conditions), **Syndromes**.

3. Syndactyly

- **Commonest congenital hand anomaly (about 1 in 2,000-2,500 live births)**; **boys 2:1**; **bilateral about 50%**; **familial (autosomal dominant with variable penetrance) in about 10-40%**.
- **Commonest site: third web space (middle-ring fingers) (about 50%)**, then 4th, 2nd, 1st (rare).
- **Classification:**
  - **Simple** (skin/soft tissue only) vs **complex** (**bony fusion: side-to-side, usually distal phalanges**),
  - **Complete** (extends to the fingertip) vs **incomplete**,
  - **Complicated** (**abnormal bones: e.g. Apert syndrome, synpolydactyly**).
- **Associated syndromes:**
  - **Apert syndrome (acrocephalosyndactyly; FGFR2 mutation)**: **craniosynostosis, mid-face hypoplasia, complex "mitten hand" syndactyly (rosebud hand)**, **symphalangism**, **radial clinodactyly of the thumb (delta phalanx)**,
  - **Poland syndrome**: **unilateral absence of the sternocostal head of pectoralis major**, **ipsilateral symbrachydactyly (short webbed fingers: hypoplastic middle phalanges)**, **breast/nipple hypoplasia**,
  - **Constriction band syndrome (acrosyndactyly: fenestrated distal fusion with proximal sinus tracts)**,
  - **Carpenter syndrome**, **Pfeiffer syndrome**.

- **Timing of release:**
  - **Border digits (thumb-index, ring-little) first and earlier (about 6-12 months)**: **unequal lengths cause progressive tethering/angular deformity of the longer digit**,
  - **Central digits (middle-ring): about 12-18 months** (before school age),
  - **Release only one side of a digit at a time** (**never both sides of a finger simultaneously: risk of vascular compromise**),
  - Complex syndactyly: earlier staged release.
- **Technique: dorsal rectangular or pentagonal flap to reconstruct the web commissure** (**the commissure slopes from dorsal-proximal to palmar-distal at about 45 degrees**), **interdigitating zig-zag flaps**, **full-thickness skin grafts (groin/antecubital)** for skin deficit (**syndactylised digits share skin: almost always need grafts**); **Buck-Gramcko pulp plasty** for fingertip separation (nail folds).
- **Complications: web creep (commonest: distal migration of the web: especially with split-thickness grafts or early surgery)**, **flexion contractures (scar)**, **graft loss**, **nail deformity**, **vascular compromise**, **angular deformity**.

4. Polydactyly

- **Postaxial (ulnar) polydactyly: commonest overall** (**about 10x more common in individuals of African descent**; **autosomal dominant**); **Temtamy and McKusick: type A (well-formed digit articulating with the fifth/sixth metacarpal)**, **type B (rudimentary, pedunculated skin tag ("nubbin"))**. **In Caucasians, postaxial polydactyly suggests a syndrome** (e.g. **Ellis-van Creveld (chondroectodermal dysplasia)**, **Bardet-Biedl**, **trisomy 13**): **genetic evaluation**. **Type B: suture ligation in the nursery (risk of a residual bump/neuroma) or surgical excision**; **type A: surgical excision with reconstruction of the collateral ligament/hypothenar muscle (ADM) insertion** at about **9-12 months**.
- **Preaxial (radial) polydactyly = thumb duplication:**
  - **Commonest in Asians and Caucasians among radial anomalies**; **usually sporadic, unilateral**.

  - **Wassel classification (level of duplication; odd numbers = bifid (partial), even numbers = complete duplication):**
  - **I** bifid distal phalanx,
  - **II** duplicated distal phalanx,
  - **III** bifid proximal phalanx,
  - **IV duplicated proximal phalanx (commonest: about 45%)**,
  - **V** bifid metacarpal,
  - **VI** duplicated metacarpal,
  - **VII triphalangeal thumb component** (associated with **Holt-Oram syndrome, Fanconi anaemia (VII + radial deficiency)**, autosomal dominant triphalangeal thumb).
  - **Wassel-Flatt** (modified) and **Rotterdam classification** (more inclusive).
  - **Treatment (about 12 months): excise the hypoplastic (usually radial) thumb**, **preserve and reconstruct the radial collateral ligament** (**reattach to the retained thumb with a periosteal sleeve**), **reattach the thenar muscles (APB) to the retained thumb**, **correct the axis (osteotomy)** and **centralise the tendons (FPL/EPL eccentric insertions)**; **Bilhaut-Cloquet procedure** (**central wedge resection and combination of both halves**: equal-sized hypoplastic thumbs: Wassel I/II; risk of **nail deformity and physeal mismatch/stiffness**).
  - **Complications: Z-deformity (zigzag: angular deformity at the IPJ/MCPJ)** (commonest late problem: from unbalanced tendons/collateral ligaments), **instability**, **stiffness**, residual nail deformity.
- **Central polydactyly** (index, middle, ring duplications): often with **syndactyly (synpolydactyly: HOXD13)**; complex reconstruction.
- **Mirror hand (ulnar dimelia)**: **duplication of ulnar side (two ulnae, no radius, multiple fingers (7-8), no thumb)**: **ZPA duplication (SHH)**; **pollicisation of one digit + excision**.

5. Radial longitudinal deficiency (radial club hand)

- **Spectrum** of hypoplasia of the radius and pre-axial (thumb) structures: **about 1 in 30,000-100,000**; **bilateral about 50%**; **thumb always involved**

(hypoplasia/absence).

- **Bayne and Klug classification (radius):**
  - **Type 0: normal radius length** (carpal deficiency/thumb hypoplasia only) (added later),
  - **Type I: short distal radius** (delayed distal radial physis),
  - **Type II: hypoplastic radius** (miniature, both physes),
  - **Type III: partial absence** (usually distal: commonest),
  - **Type IV: complete absence of the radius (commonest overall: about 50%)**.
  - (**James/Manske modified: type N** (thumb only), **type 0** (carpal anomalies with normal radius length)).
- **Associated syndromes (screen all patients):**
  - **VACTERL** (Vertebral, Anal atresia, Cardiac, TracheoEsophageal fistula, Renal, Limb anomalies),
  - **Holt-Oram syndrome** (**cardiac septal defects: ASD/VSD; TBX5, autosomal dominant**),
  - **TAR syndrome** (**thrombocytopenia-absent radius**: **thumbs PRESENT** (unique); platelets improve after the first year; autosomal recessive (RBM8A)),
  - **Fanconi anaemia** (**aplastic anaemia (develops about age 6-7: chromosomal breakage test: DEB/mitomycin C)**; needs **bone marrow transplant**; screening is essential),
  - **Trisomy 13/18**.
  - **Investigations: CBC/platelets, echocardiogram, renal ultrasound, spine radiographs, chromosomal breakage test (Fanconi)**.
- **Clinical: radially deviated, short forearm**, **ulna bowed (towards the radius)**, **thumb hypoplasia/absence**, **stiff elbow (elbow extension contracture: contraindicates centralisation, as the radially deviated wrist allows the hand to reach the mouth)**; **radial nerve absence distal to the elbow (superficial radial nerve replaced by the median nerve)**; **radial artery absent**.
- **Treatment:**
  - **Early: stretching and splinting** (from birth).

- **Types I-II (mild)**: stretching/splinting; **radial lengthening**.
- **Types III-IV**: **centralisation (placing the carpus on the distal ulna)** or **radialisation (Buck-Gramcko: carpus over-corrected onto the ulna with transfer of radial wrist extensors to the ulnar side)** at **about 6-12 months** (after soft-tissue distraction with an external fixator), followed by **pollicisation (of the index finger) for an absent/non-reconstructable thumb**; **ulnar osteotomy** for bowing.
- **Contraindications to centralisation: stiff elbow (extension contracture: patient needs radial deviation to reach the mouth)**, **adults who have adapted well**, **severe associated medical conditions (short life expectancy)**.
- **Recurrence of radial deviation (commonest complication)**, **ulnar physeal damage (growth arrest)**, stiffness.

6. Thumb hypoplasia

- **Blauth classification (modified by Manske):**
  - **Type I: minor generalised hypoplasia** (all elements present, small): no treatment,
  - **Type II: thenar muscle hypoplasia, first web space narrowing, ulnar collateral ligament (UCL) instability of the MCPJ**: **opponensplasty (Huber: ADM transfer; or FDS ring)**, **first web deepening (4-flap Z-plasty)**, **UCL reconstruction**,
  - **Type III: extrinsic tendon abnormalities (pollex abductus: FPL connection to EPL)**, **hypoplastic metacarpal**: **IIIA: stable CMCJ (basal joint present)**: **reconstruct (like type II + extrinsic tendon correction)**; **IIIB: unstable CMCJ (absent base of metacarpal)**: **pollicisation** (Manske: **critical distinction: presence of a stable CMCJ**),
  - **Type IV: pouce flottant ("floating thumb"): rudimentary thumb attached by a skin bridge**: **pollicisation**,
  - **Type V: absent thumb**: **pollicisation**.
- **Pollicisation (Buck-Gramcko technique)**: **index finger transferred to the thumb position**: **rotated about 120-160 degrees (pronation)**, **abducted about 40 degrees**, **shortened** (**index MCPJ becomes the new**

CMCJ; metacarpal head becomes the trapezium; PIPJ becomes the MCPJ); **first dorsal interosseous becomes APB/abductor**, **first palmar interosseous becomes adductor**, **EIP becomes EPL**, **EDC (index) becomes APL**; **at about 12-18 months**.

7. Ulnar longitudinal deficiency

- **Less common than radial (about 1:4-10 relative frequency)**; **not associated with systemic syndromes** (unlike radial deficiency) but **associated with other musculoskeletal anomalies (PFFD, fibular hemimelia, scoliosis)** and **hand anomalies (ulnar digits absent, syndactyly)**.
- **Bayne classification: I hypoplastic ulna**, **II partial aplasia**, **III total aplasia**, **IV radiohumeral synostosis**.
- **Elbow instability or synostosis** (vs radial deficiency: wrist is relatively stable).
- **Treatment:** focus on **hand function (syndactyly release, thumb reconstruction, first web)**; **excision of the ulnar anlage** (fibrocartilaginous band that tethers the radius: progressive bowing/radial head dislocation), **one-bone forearm** for severe instability.

8. Central deficiency (cleft hand) vs symbrachydactyly

Feature	Cleft hand (typical; central longitudinal deficiency)	Symbrachydactyly (atypical cleft)
Shape	**V-shaped central cleft** (missing middle ray)	**U-shaped**; **short digits with nubbins (rudimentary nails)**
Inheritance	**Autosomal dominant; often bilateral**; **feet involved (split hand-foot malformation: SHFM)**	**Sporadic, unilateral**
Associations	**EEC syndrome (ectrodactyly-ectodermal dysplasia-clefting)**, syndactyly of the first web	**Poland syndrome**

Feature	Cleft hand (typical; central longitudinal deficiency)	Symbrachydactyly (atypical cleft)
Mechanism	**AER (central) failure**	**AER/mesoderm (proximal-distal) failure: transverse spectrum**
Function	**Often excellent ("functional triumph, aesthetic disaster" (Flatt))**	Variable
Treatment	**Cleft closure (Snow-Littler, Ueba, Miura)** + **first web release** (**syndactyly of the first web: release first**); **transposition of the index ray** to the third metacarpal position	**Non-vascularised toe phalanx transfer** (**under 18 months: physis survives**), **free toe transfer**, **distraction lengthening**, **web deepening**

9. Camptodactyly and clinodactyly

Camptodactyly

- **Non-traumatic flexion contracture of the PIPJ** (**little finger commonest**; **bilateral about 2/3**); **infantile (equal sexes)** and **adolescent (girls; progressive during growth spurt)** forms.
- **Cause: abnormal lumbrical insertion (commonest)**, **FDS contracture/abnormal origin**, **skin deficiency**, volar plate contracture, **abnormal insertion of the lumbrical/interosseous into the FDS or capsule**.
- **Radiographs (late): flattening (notching) of the neck of the proximal phalanx, flattened/indented base of the middle phalanx, volar subluxation**.
- **Treatment: splinting and stretching (mainstay; most effective under about 3 years and in flexible deformity)**; **surgery (FDS tenotomy/lengthening, abnormal lumbrical release, volar plate release, skin flaps/grafts)** for **contractures over 30-60 degrees with functional impairment or progression despite splinting**; **outcomes are unpredictable** (stiffness, loss of flexion).

Clinodactyly

- **Angular deformity in the radioulnar (coronal) plane**, **little finger (radial deviation at the DIPJ) commonest**; **middle phalanx**: **trapezoidal or triangular (delta phalanx: longitudinal epiphyseal bracket (C-shaped physis))**.
- **Associations: Down syndrome (about 35-80%)**, Klinefelter, Turner, **Rubinstein-Taybi (broad thumbs, delta phalanx)**, **Apert**.
- **Treatment: observation (most: cosmetic)**; **surgery if over 30 degrees with functional impairment/overlapping**: **opening-wedge, closing-wedge or reverse-wedge osteotomy**; **physiolysis (Vickers: excision of the longitudinal bracket with fat interposition)** in **young children (under 6 years: allows growth to correct)**.
- **Kirner deformity**: **palmar-radial curvature of the distal phalanx of the little finger** (**girls 8-14 years**): observation/osteotomy.

10. Other conditions

- **Trigger thumb (paediatric): not truly congenital (develops after birth; usually noticed at 1-3 years)**; **fixed flexion of the IPJ** with a **Notta node (nodule in the FPL at the A1 pulley)**; **about 30% resolve spontaneously (more under 1 year)**; **A1 pulley release** if persistent beyond about **3 years** (or earlier if fixed) (**the radial digital nerve is close and superficial: at risk**).
- **Trigger finger (paediatric):** rarer; **anatomical abnormalities (FDS decussation, abnormal slips, nodules)**: **A1 release alone is often insufficient: excise one slip of FDS ± A3 release**.
- **Congenital clasped thumb: absent/hypoplastic EPB/EPL** (thumb flexed across the palm); **splinting (3-6 months)**; **tendon transfer (EIP to EPL)** and **web reconstruction** if it fails; associated with **arthrogryposis** (**complex clasped thumb**).
- **Constriction ring (amniotic band) syndrome: circumferential constriction rings**, **acrosyndactyly (fenestrated distal syndactyly)**, **amputations**, **lymphoedema distal to the ring**; **not genetic (sporadic)**; **urgent release for vascular compromise (neonatal emergency)**;

staged Z-plasty/W-plasty release (half the circumference at a time traditionally; single-stage now common).

- **Macrodactyly: true overgrowth of all tissues of a digit** (**PIK3CA-related overgrowth spectrum (PROS)**; **lipofibromatous hamartoma of the median nerve** ("nerve territory-oriented macrodactyly": index/middle)); **static vs progressive**; **epiphysiodesis, debulking, ray amputation**; differentials: **Proteus syndrome**, **neurofibromatosis**, **Klippel-Trenaunay**, **Ollier/Maffucci**.
- **Arthrogryposis multiplex congenita (amyoplasia commonest)**: **internally rotated shoulders, extended elbows, flexed and ulnarly deviated wrists, thumb-in-palm**, **symmetrical limb involvement, absent skin creases**, **normal intelligence and sensation**; **early stretching/serial casting**; **elbow flexion (passive flexion first: posterior capsulotomy + triceps lengthening; then tendon transfers (triceps to biceps, pectoralis, latissimus))**; **wrist: dorsal closing-wedge carpal osteotomy**.
- **Congenital radioulnar synostosis: proximal (commonest)**, **fixed pronation**, **often bilateral**; **AD inheritance in some; associated with sex chromosome anomalies (XXY)**; **no rotation (no supination)**; **observation if pronation under about 60 degrees** (**compensated by shoulder/wrist**); **derotational osteotomy through the synostosis to about 10-20 degrees pronation (dominant) and neutral/slight supination (non-dominant)** if severe (over about 60 degrees); **compartment syndrome and vascular compromise risk with large rotations**; **resection of the synostosis fails (re-forms)**.
- **Madelung deformity: volar-ulnar tethering of the distal radial physis (Vickers ligament: abnormal radiolunate ligament)**: **volar and ulnar tilt of the distal radius, dorsally prominent ulna**, **adolescent girls, bilateral**; **Léri-Weill dyschondrosteosis (SHOX gene)**; **Vickers ligament release with physiolysis (younger)**, **dome osteotomy of the radius ± ulnar shortening** (older).
- **Congenital radial head dislocation: usually posterior**, bilateral, **convex radial head, hypoplastic capitellum** (vs traumatic: concave head, normal capitellum); **associated with syndromes (nail-patella, Ehlers-Danlos)**; **observe; excise the radial head at maturity** if painful.

Exam pearl: **AER (FGF) = proximal-distal (transverse deficiency); ZPA (SHH) = radial-ulnar (polydactyly, mirror hand); Wnt-7a/LMX1B = dorsal-ventral (nail-patella). OMT classification replaced Swanson.**

Exam pearl: **Syndactyly: commonest; 3rd web; border digits first (6-12 months); never both sides of one digit; dorsal flap commissure + FTSG; web creep. Apert (FGFR2) = complex mitten hand; Poland = pectoralis head + symbrachydactyly.**

Exam pearl: **Thumb duplication: Wassel IV commonest; excise radial thumb, keep RCL and thenar insertion; Bilhaut-Cloquet for equal halves; Z-deformity late. Postaxial polydactyly in Caucasians = syndrome (Ellis-van Creveld).**

Exam pearl: **Radial deficiency: screen for VACTERL, Holt-Oram (heart), TAR (thumbs present), Fanconi (chromosomal breakage). Bayne IV commonest. Centralisation/radialisation contraindicated with a stiff elbow. Blauth IIIA reconstruct, IIIB/IV/V pollicise (stable CMCJ is the key). Pollicisation: index rotated about 120-160 degrees; MCPJ becomes CMCJ.**

Exam pearl: **Cleft hand (V, bilateral, feet, AD) vs symbrachydactyly (U, unilateral, nubbins, Poland). Camptodactyly: PIPJ little finger (lumbrical), splint. Clinodactyly: delta phalanx, physiolysis under 6. Paediatric trigger thumb: Notta node, release after 3 years (radial digital nerve at risk). Synostosis: osteotomy, not resection. Madelung: Vickers ligament, SHOX.**